

### PRACTICAL 3: Creating Advanced Visualizations in Tableau

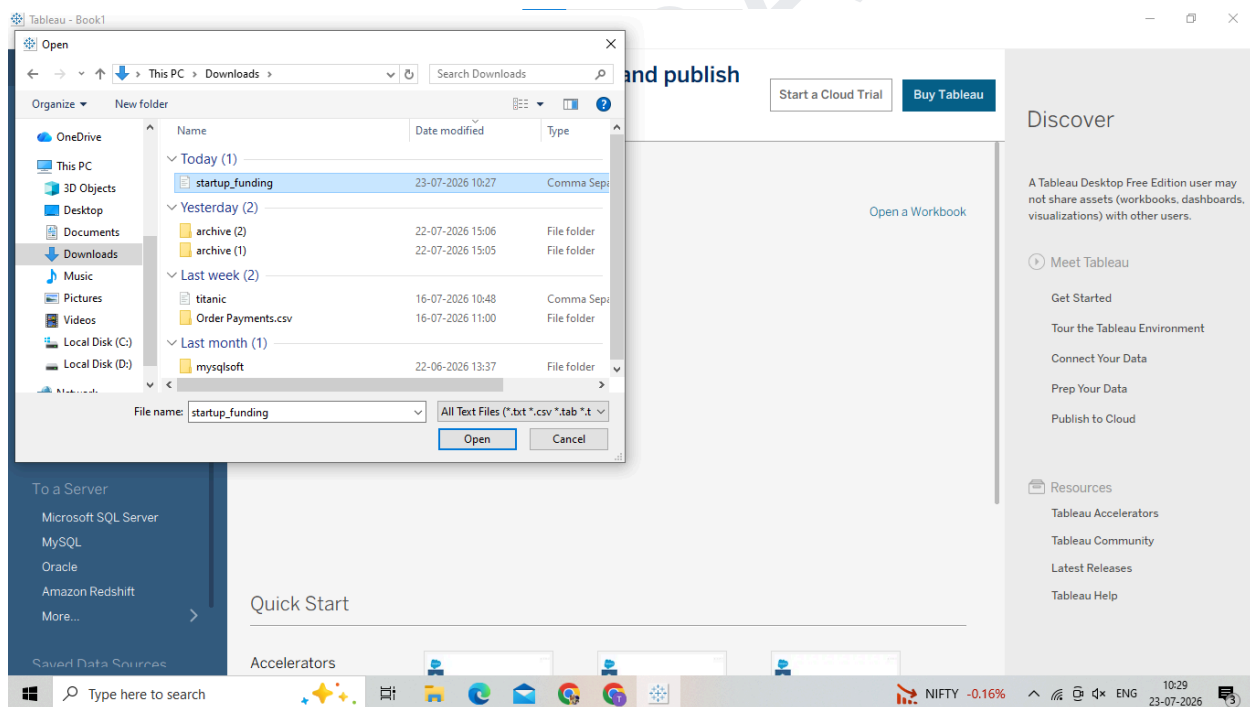
- a) Create stacked and grouped bar charts for comparison
- b) Build treemaps and pie charts for part-to-whole analysis
- c) Design scatter plots to identify relationships
- d) Create dual-axis and combination charts
- e) Develop histograms for distribution analysis
- f) Build box-and-whisker plots
- g) Use Gantt charts and advanced date visualizations

Dataset : Indian Startup Funding

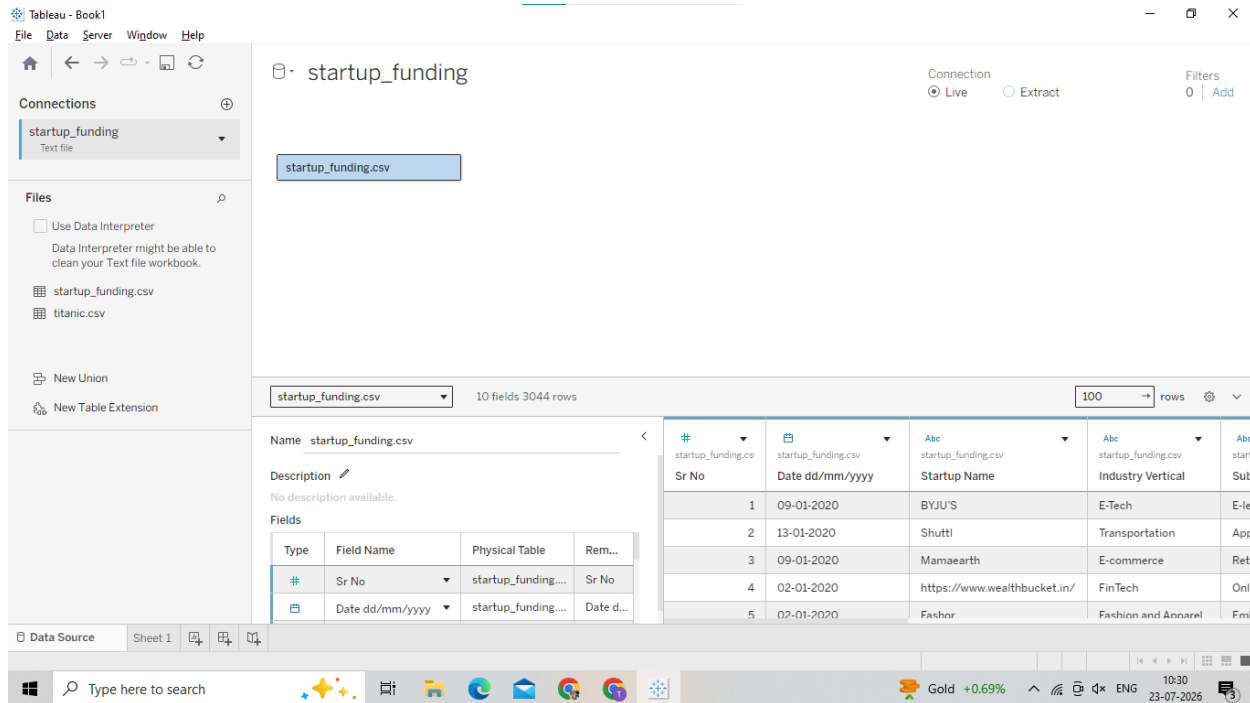
#### Step A: Creating Stacked and Grouped Bar Charts for Comparison

##### 1. Connecting to the Dataset

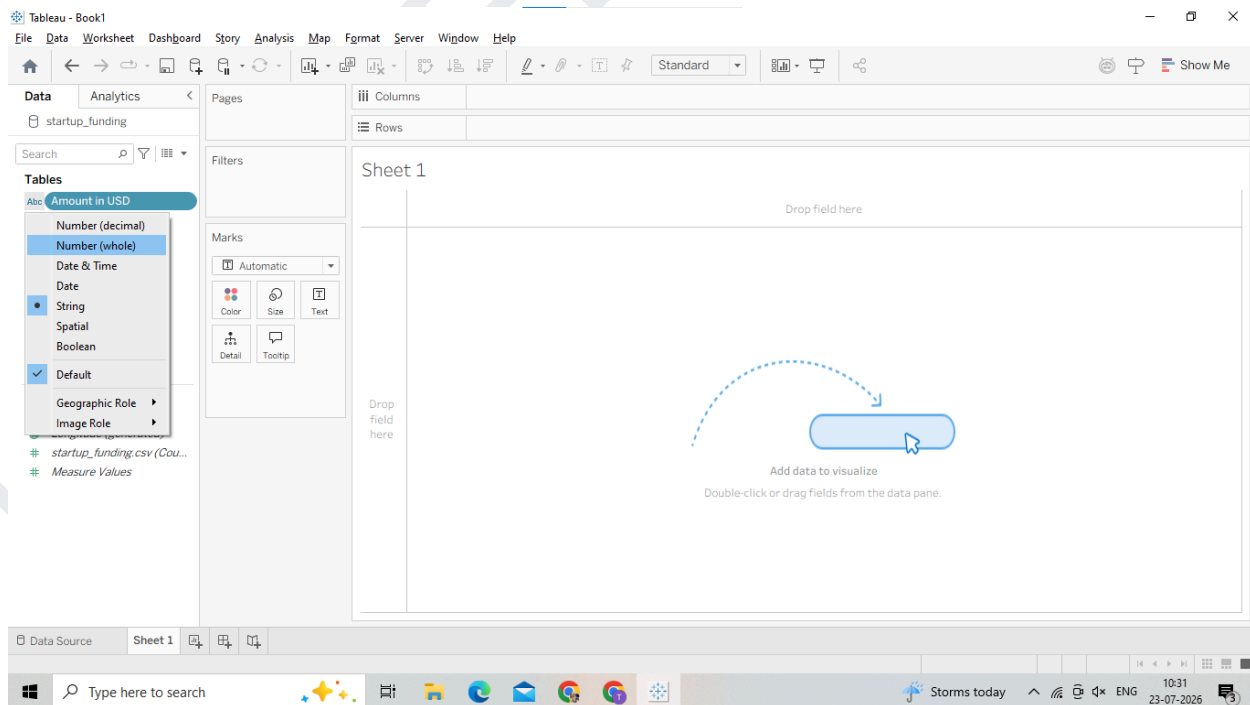
1. Launch Tableau. Under the Connect panel, click on Text File, select your startup\_funding.csv, and click Open.



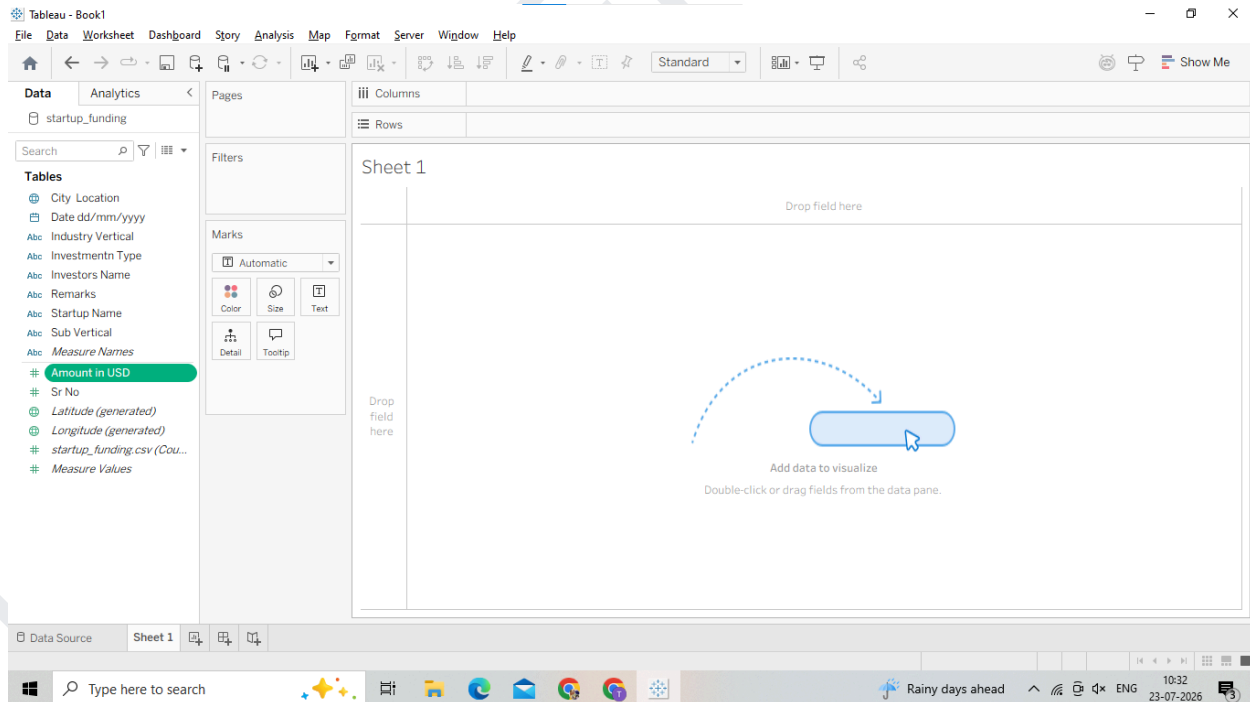
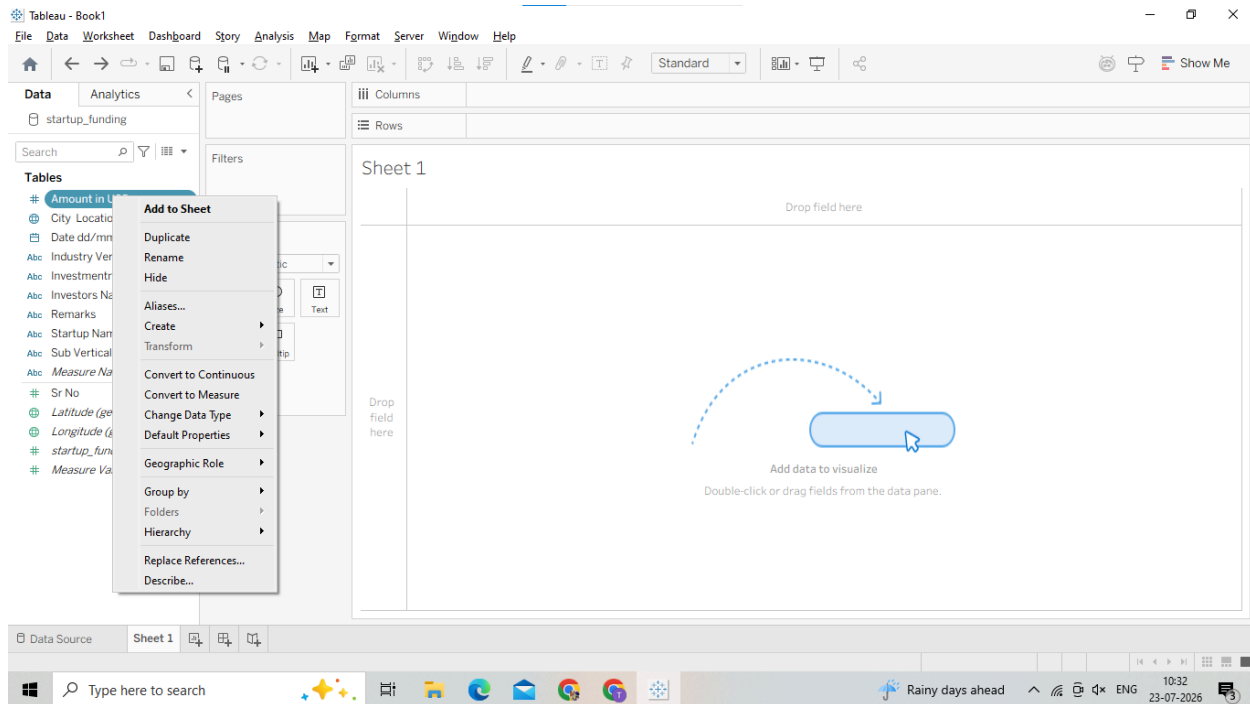
2. Click on Sheet 1 at the bottom-left corner to enter your workspace.



3. Fix the Data Type: In the Data pane on the left, locate Amount in USD. Click the small Abc text icon next to it and select Number (Whole) from the dropdown menu to convert it into numeric data.

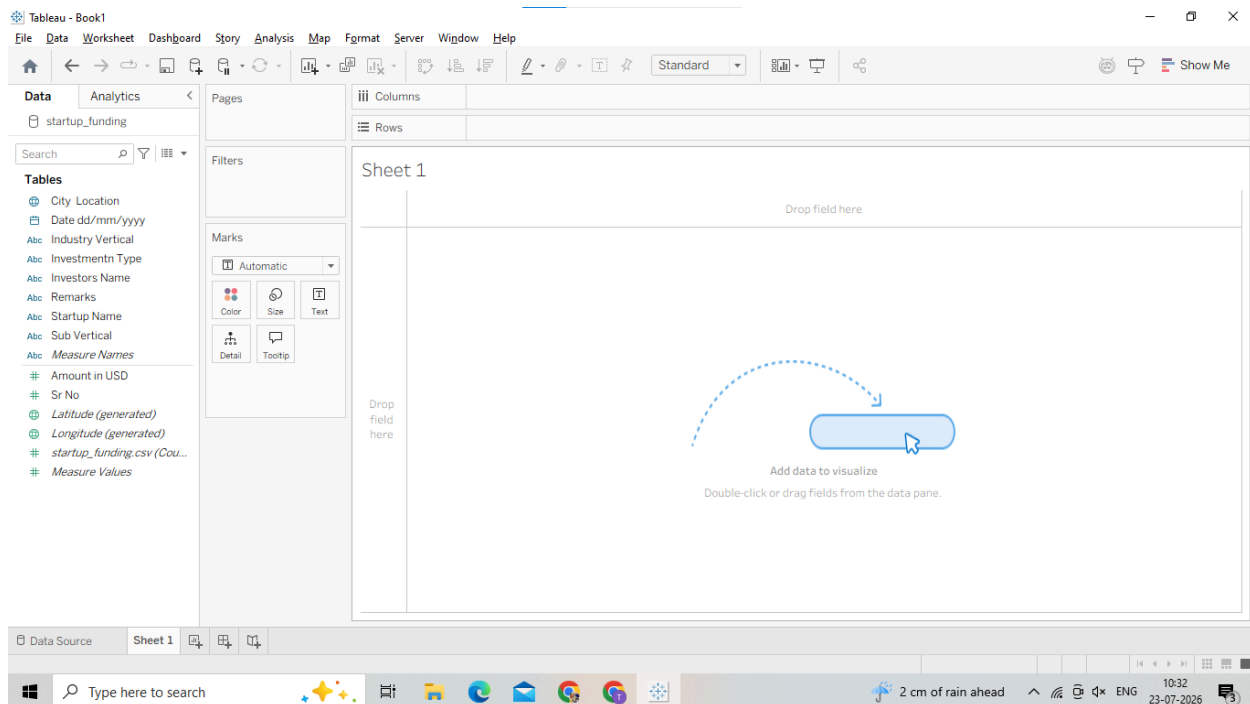


4. Convert to Measure: Right-click Amount in USD in the Data pane and select Convert to Measure. It will turn green and move to the bottom half of the pane.

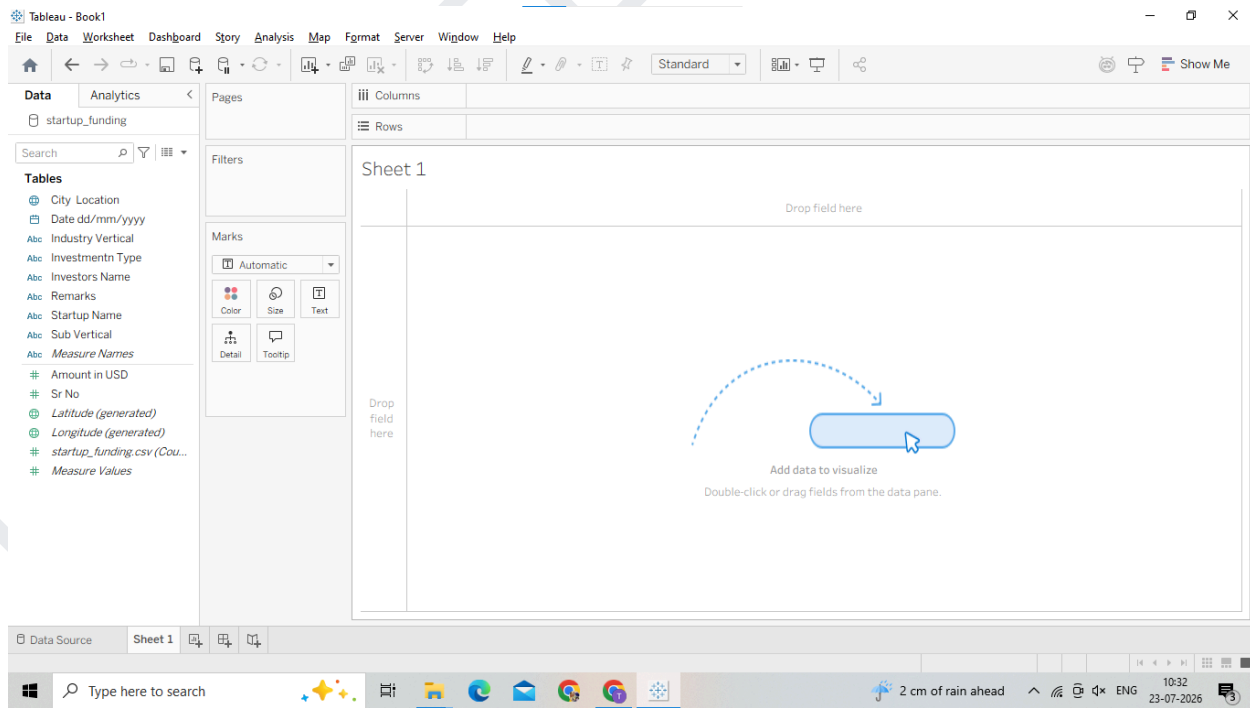


## 2. Building a Stacked Bar Chart

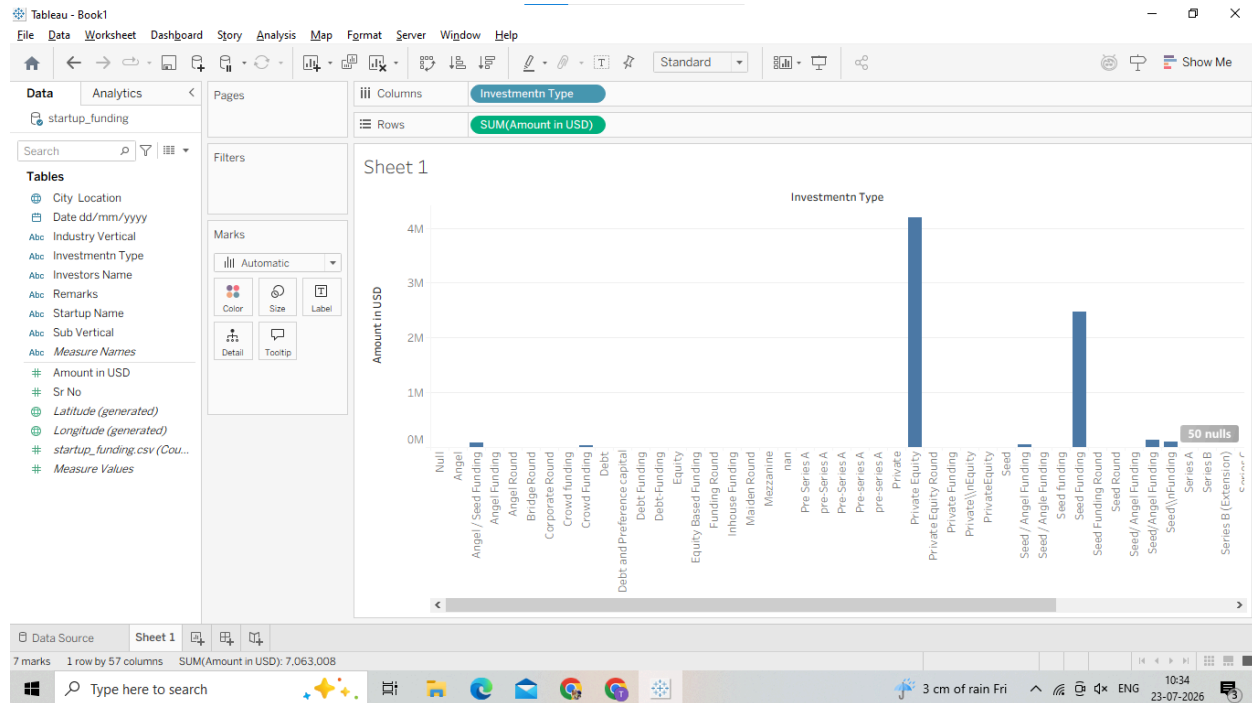
1. From the Data Pane on the left, click and drag the InvestmentnType dimension onto the Columns shelf.



2. Drag the Amount in USD measure onto the Rows shelf.



3. Take the City Location dimension from the Data Pane and drop it directly onto the Color tile in the Marks card.

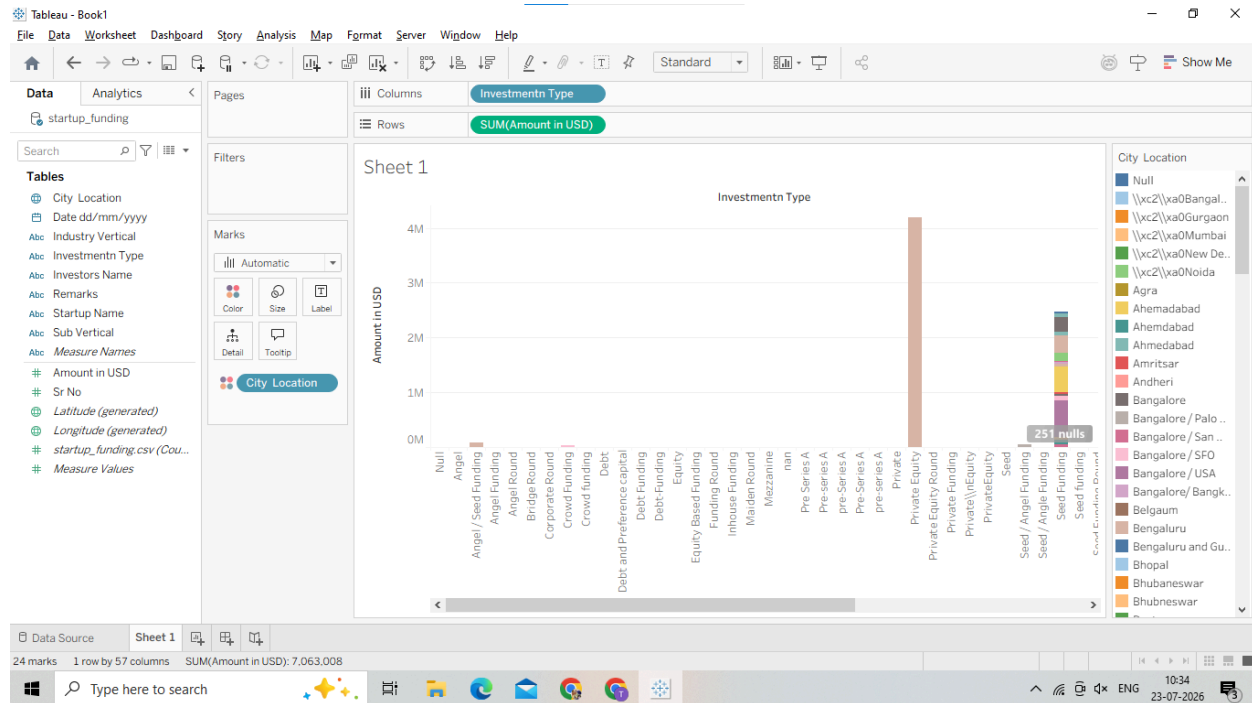


**Why we are doing it:** A **Stacked Bar Chart** places categorical data segments on top of one another within a single bar. We use this to see the total volume of funding per investment type while simultaneously breaking down which cities contributed to that Total.

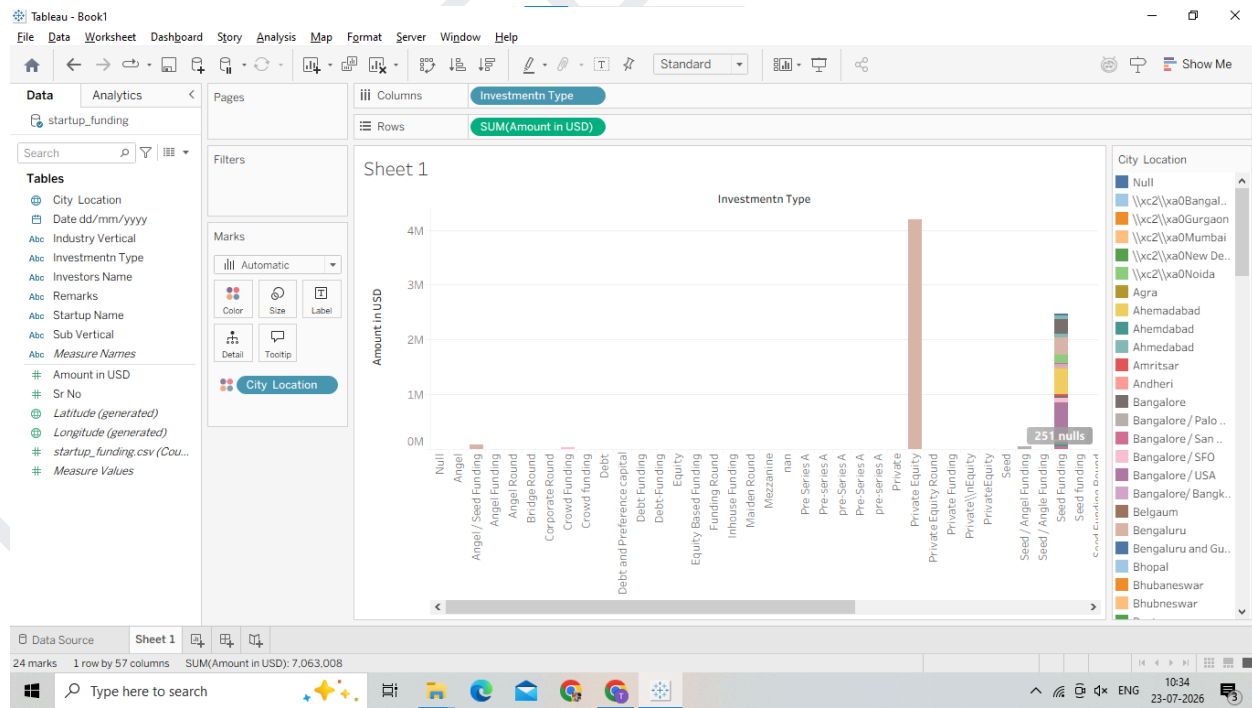
**Usage:** Use a stacked bar chart when you want to show a comparison of total values across broad categories, while highlighting how much individual sub-categories (like cities) contribute to the whole.

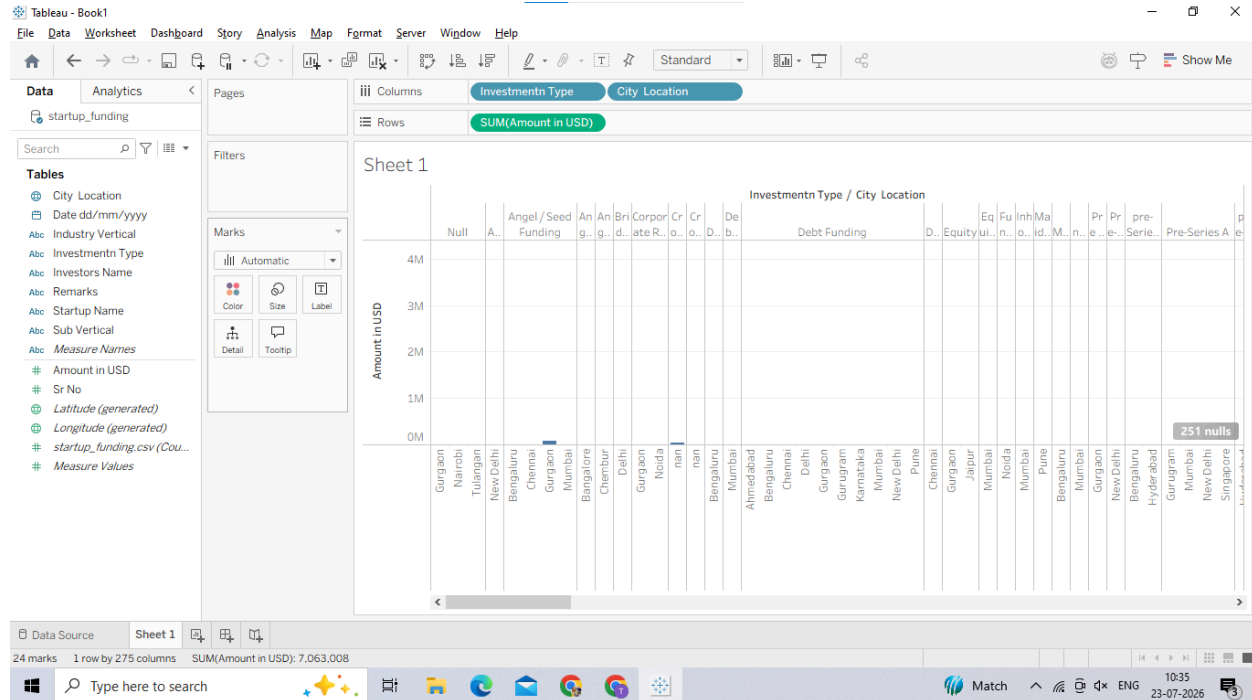
### 3. Converting to a Grouped Bar Chart (Side-by-Side)

#### 1. Leave Amount in USD on the Rows shelf.



#### 2. Take the City Location pill out of the Color tile on the Marks card and drag it up to the Columns shelf, placing it right next to (to the left of) the InvestmentnType pill.



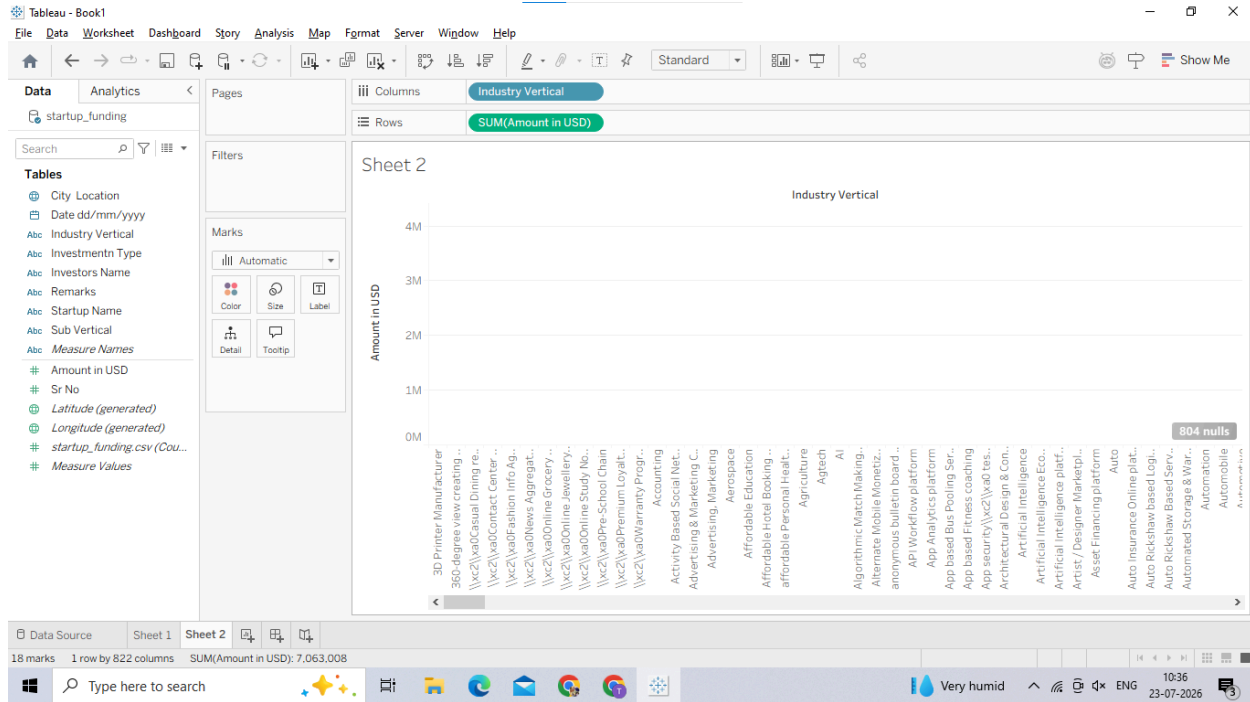


**Why we are doing it:** A Grouped (or Side-by-Side) Bar Chart rearranges the stacked segments into individual, standalone bars grouped under a main category header.

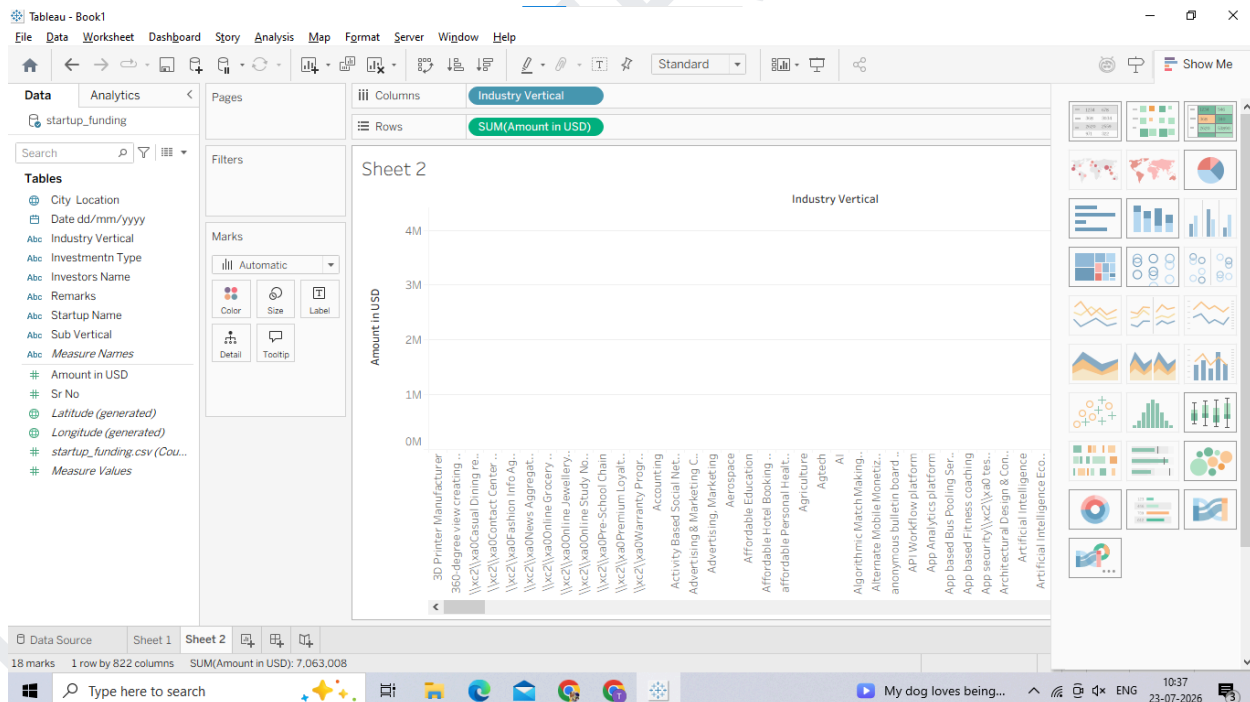
**Usage:** Use a grouped bar chart when your primary goal is to compare the exact heights of individual sub-categories directly against each other (e.g., comparing Mumbai's seed funding versus Bengaluru's seed funding side-by-side) rather than focusing on the cumulative total.



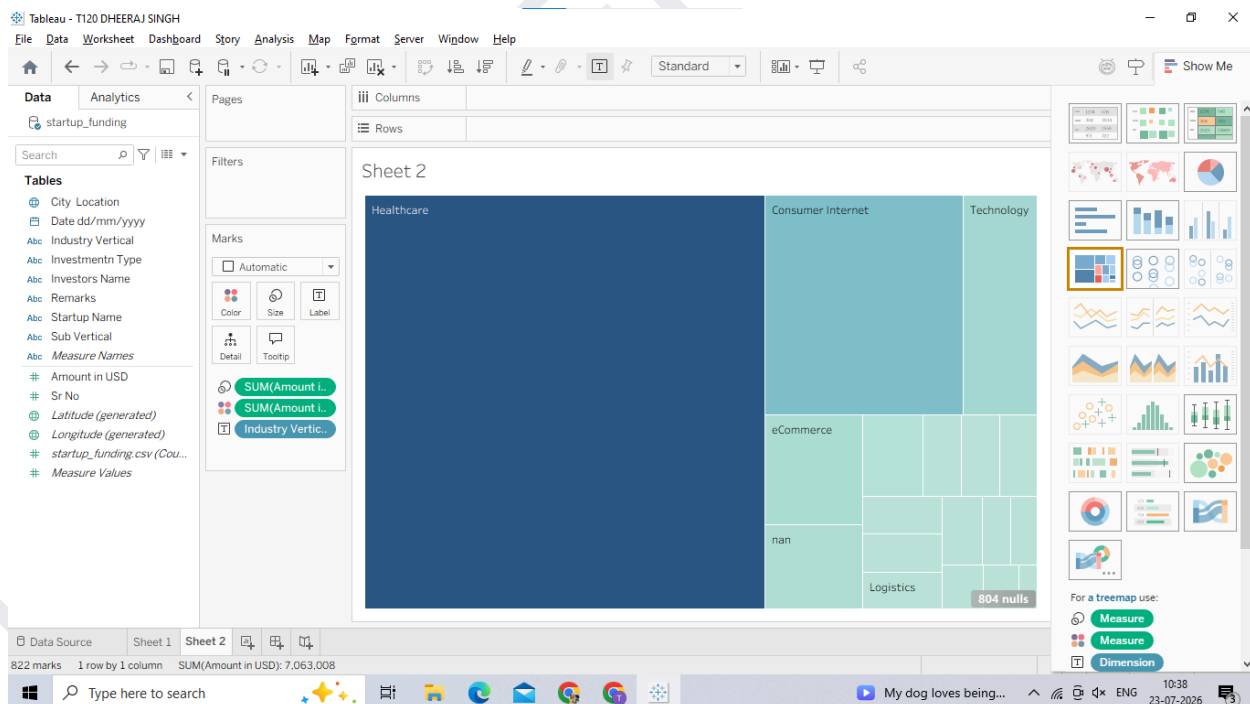
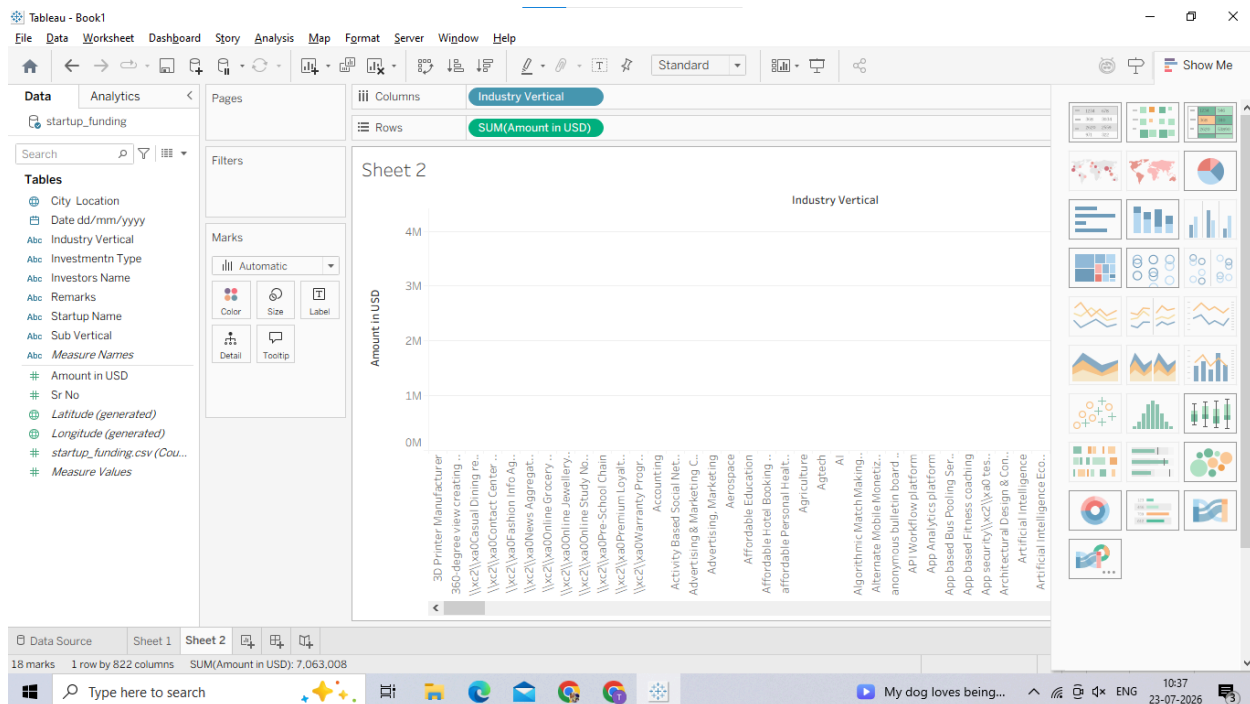
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4. Now, look at the very top-right corner of the window and click the Show Me button to expand the chart library panel.



5. From the options, click on the Treemap icon (it looks like a grid of nested colorful rectangles).



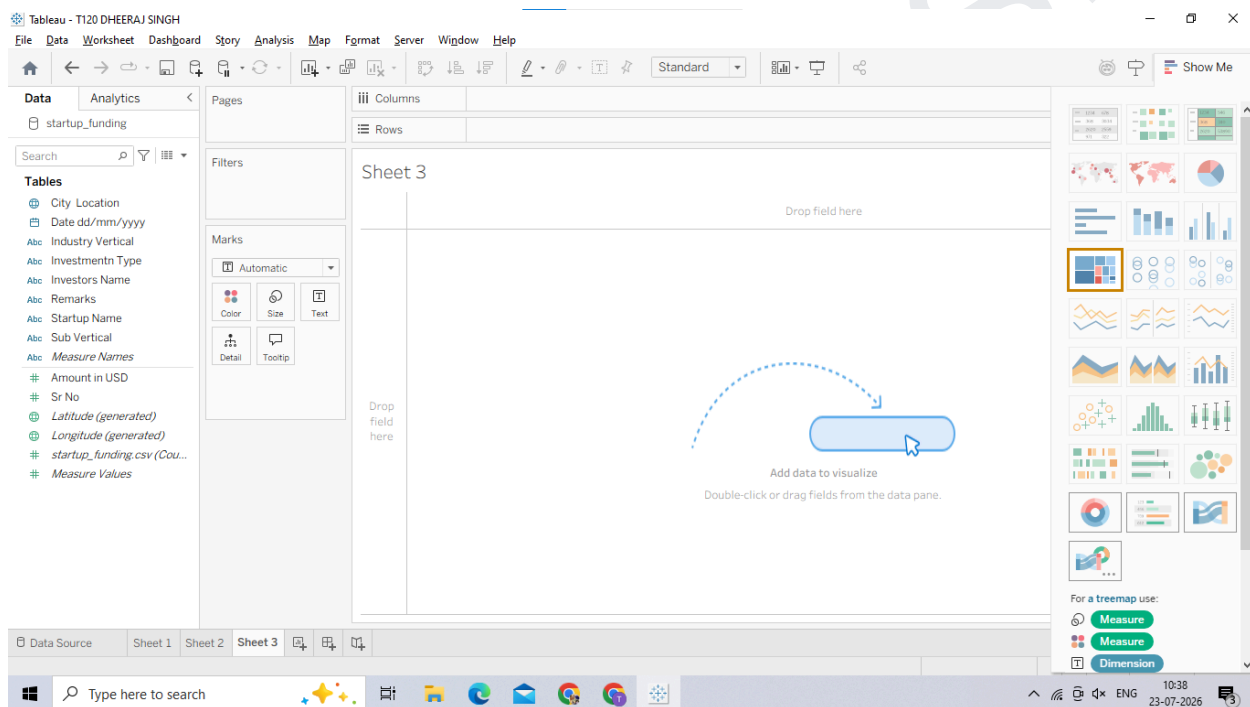
**Why we are doing it:** A Treemap displays hierarchical or categorical data using nested rectangles. The area (size) and color shade of each rectangle represent the quantitative

value. We use this to see at a glance which sector (like FinTech or E-commerce) commands the largest market share of startup capital.

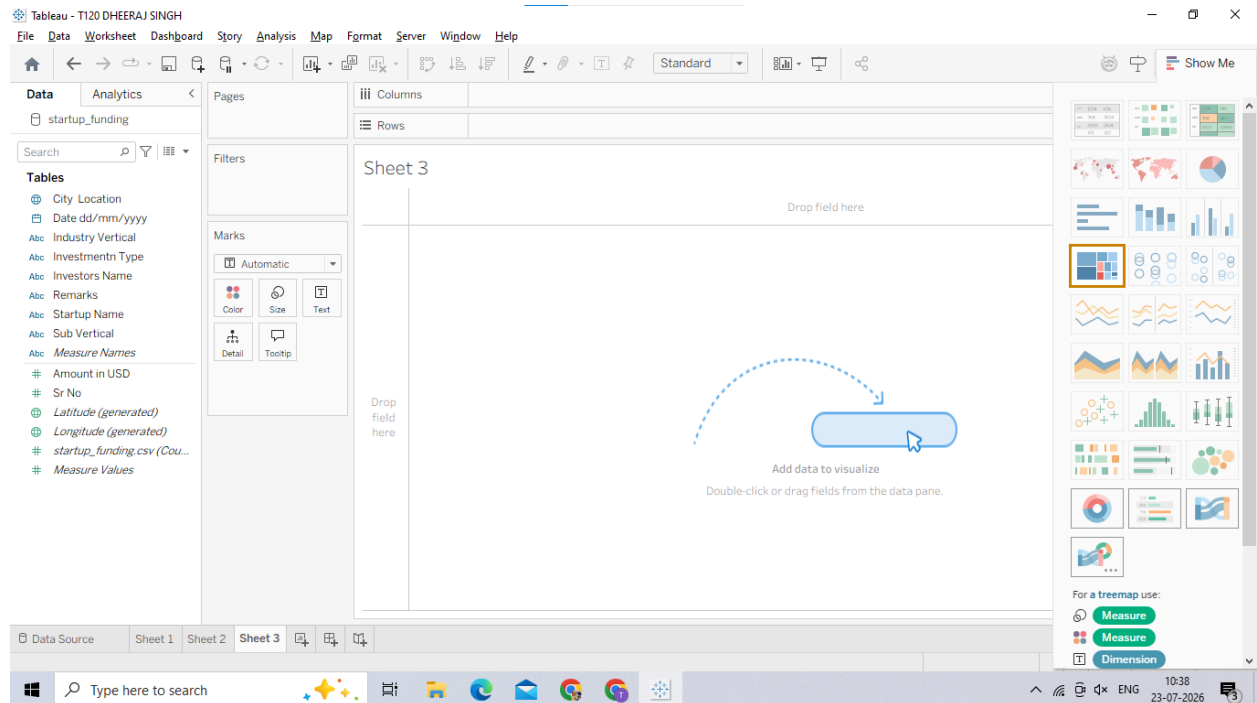
**Usage:** Use a treemap when you want to perform a part-to-whole analysis on a categorical field with many distinct variables (like dozens of industries). It handles a large volume of dimensions much better than a pie chart without looking cluttered.

## 2. Building a Pie Chart

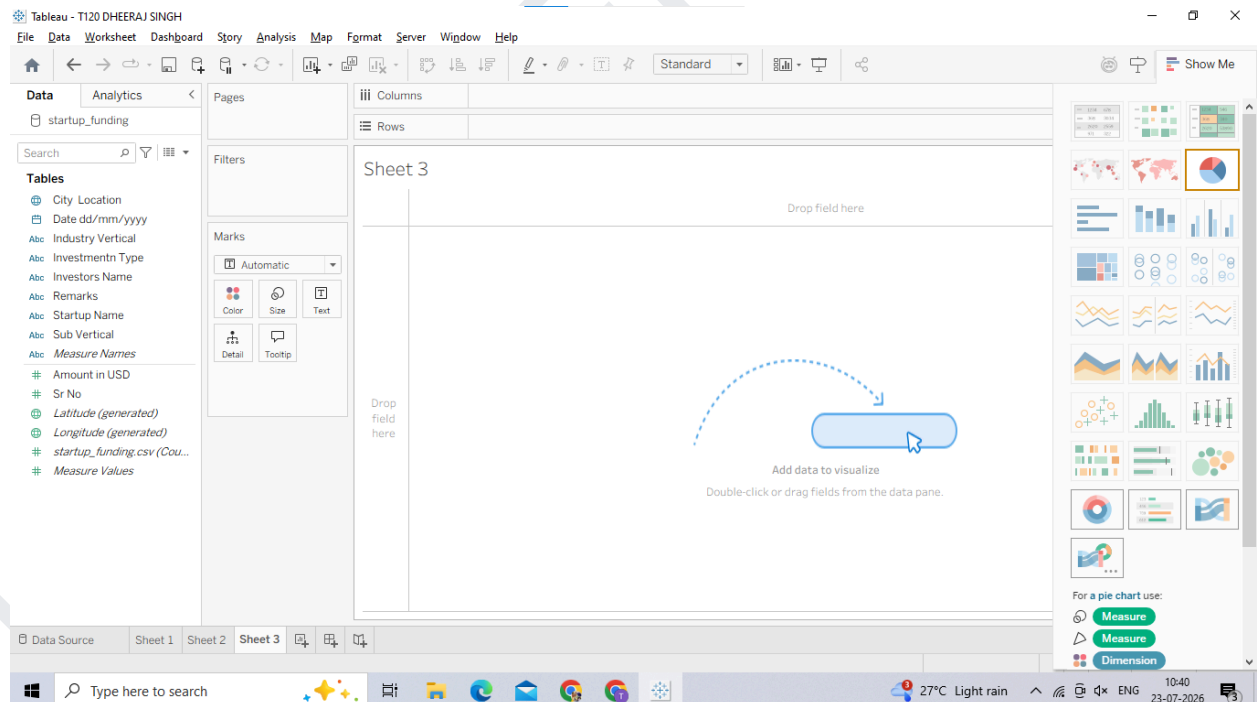
1. Click the New Worksheet icon to open another blank layout (Sheet 3).

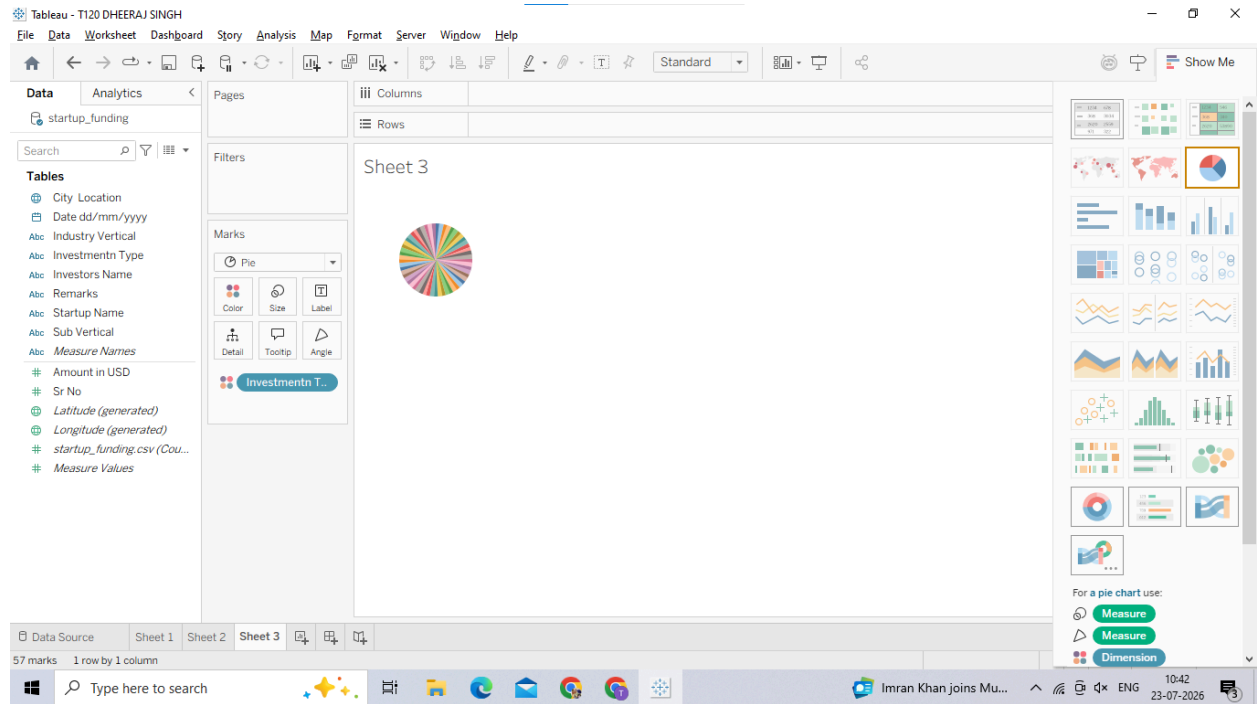


2. Go back to the Show Me panel in the top-right corner and select the Pie Chart icon.

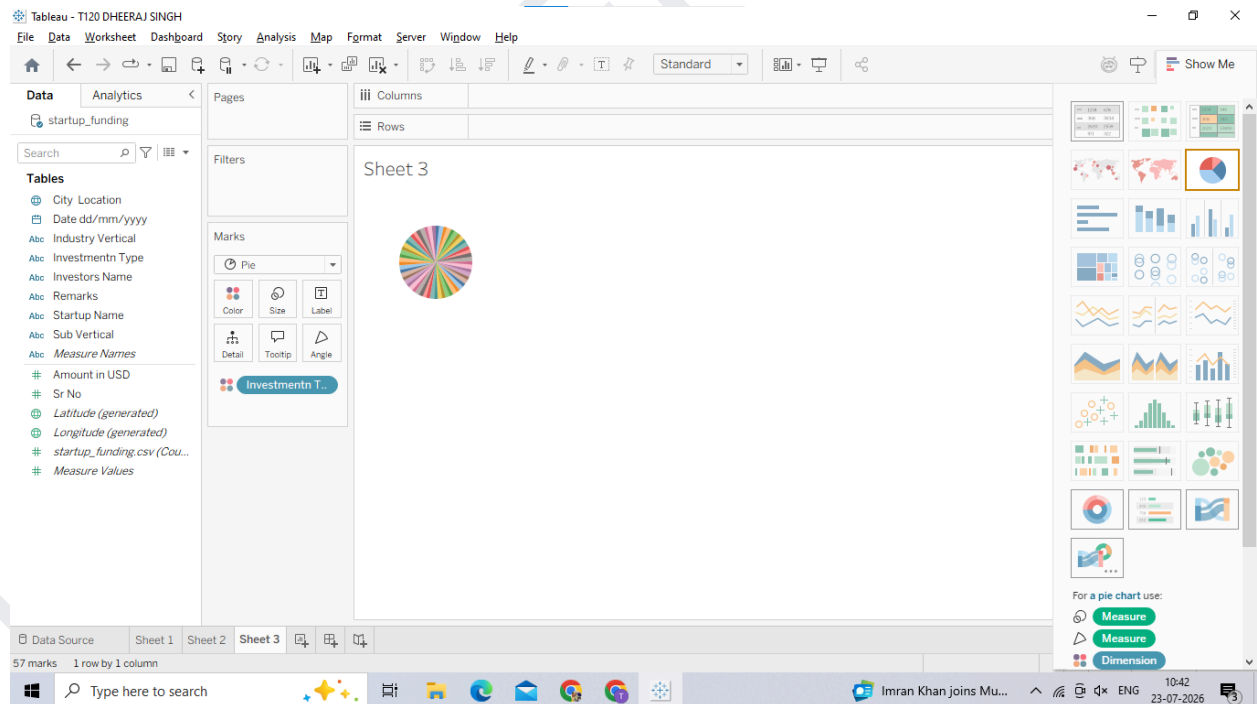


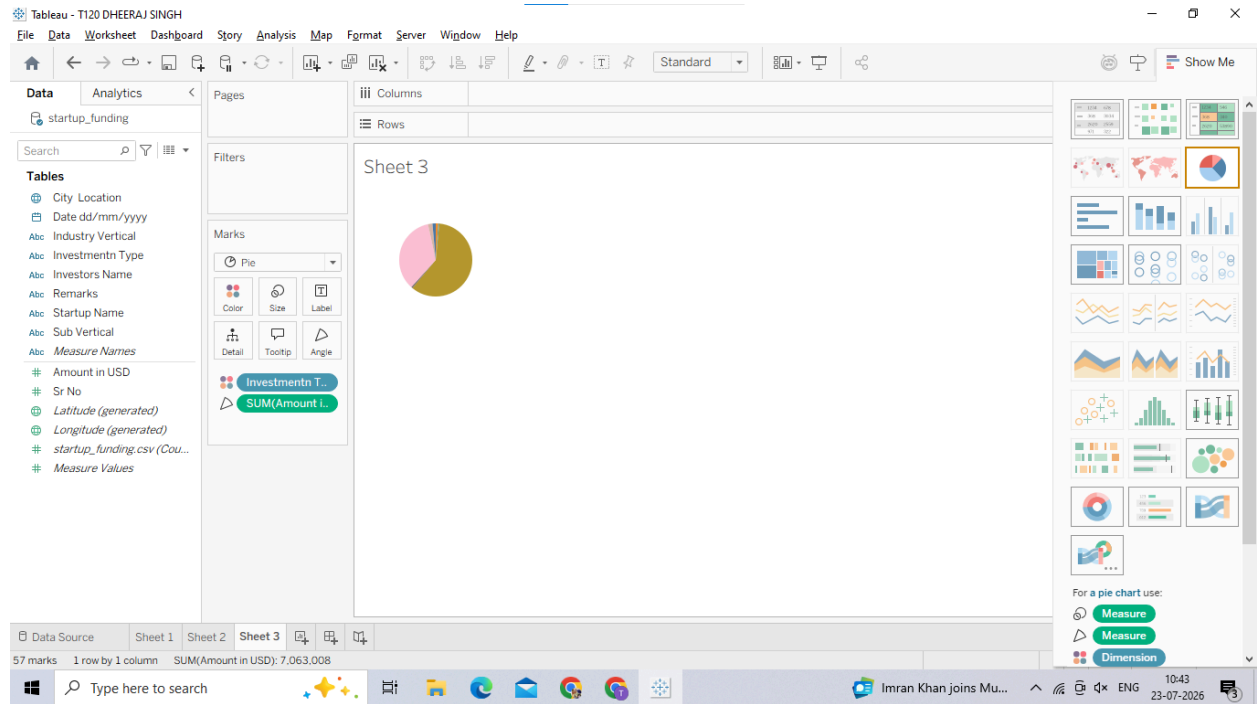
3. Notice that your Marks card changes to show an angle option. From the Data Pane, drag InvestmentnType and drop it onto the Color tile in the Marks card.



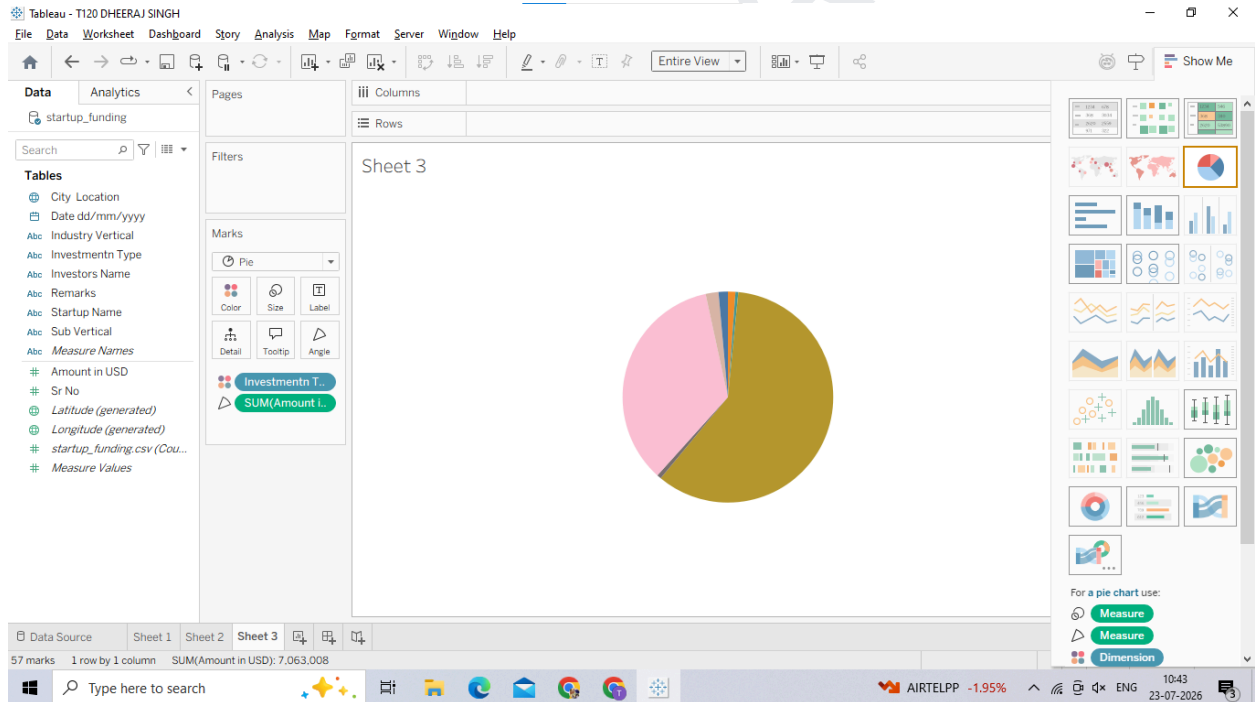
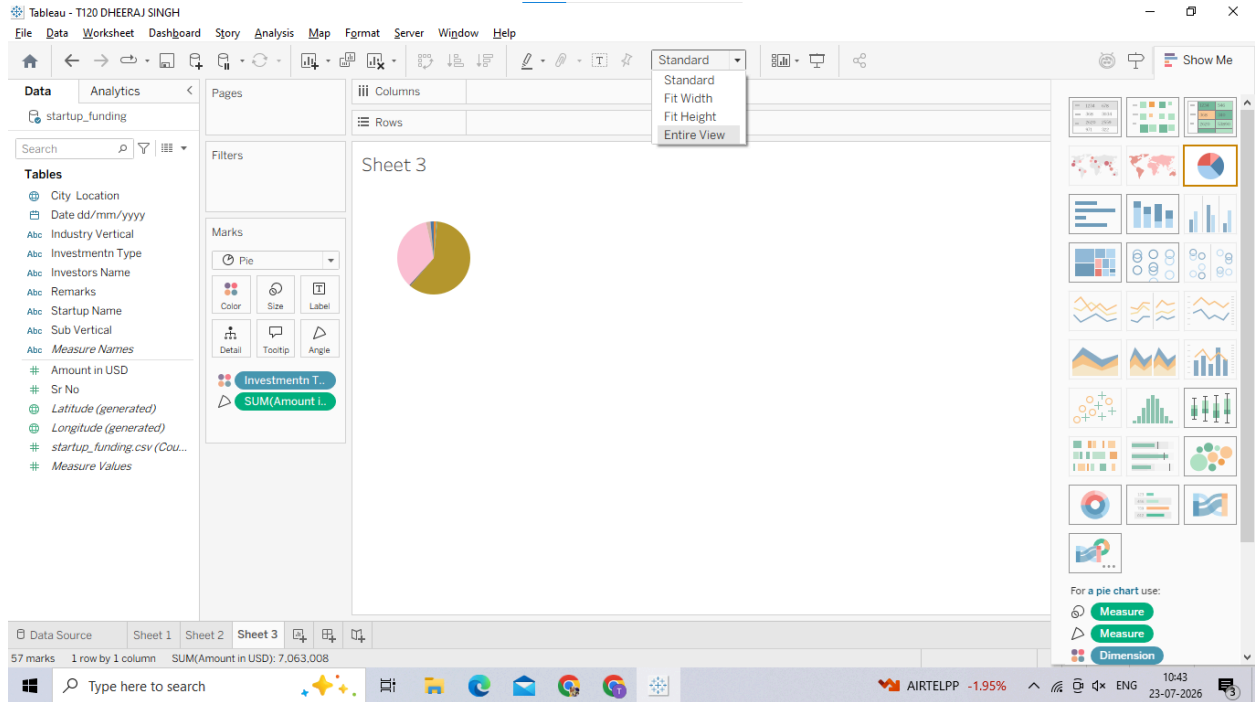


4. Drag Amount in USD from the Data Pane and drop it directly onto the Angle tile in the Marks card.





5. Enlarge the view: To make the pie chart legible, look at the top toolbar menu, find the fit dropdown menu (which says Standard by default), and switch it to Entire View.



**Why we are doing it:** A Pie Chart represents part-to-whole data slices in a circular layout. By applying InvestmentType to color and Amount in USD to angle, the size of each slice corresponds directly to its funding contribution.

**Usage:** Use a pie chart when your categorical field contains very few variables (ideally 2 to 5 options, such as comparing Seed Funding vs. Private Equity). This ensures that

viewers can quickly evaluate proportions without the slices becoming too narrow to interpret.

## Step C: Designing Scatter Plots to Identify Relationships

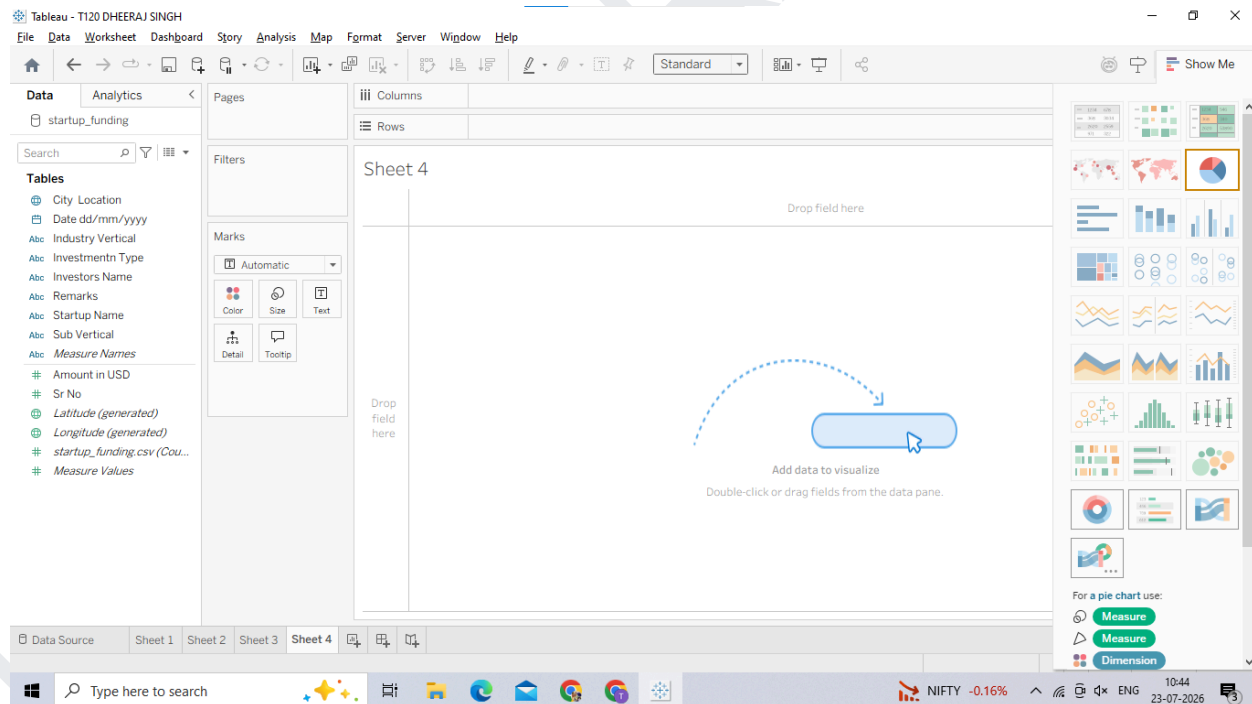
### 1. Cleaning the Data for Scatter Plots

A scatter plot maps relationships between two different numerical measures. However, in this dataset, we only have one explicit primary numeric field: Amount in USD.

To create a meaningful scatter plot, we will instruct Tableau to generate a second numerical measure dynamically by counting the number of times a startup name appears (representing the total number of funding rounds a startup went through).

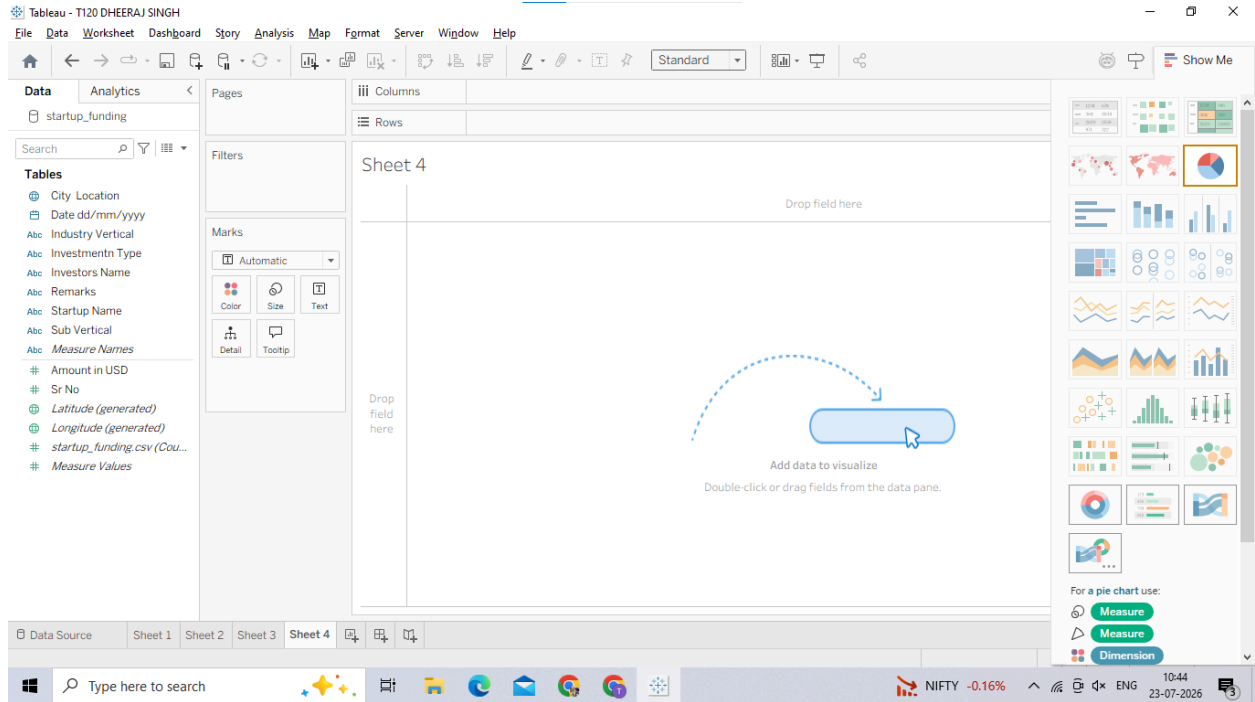
### 2. Building the Scatter Plot

1. Click the New Worksheet icon at the bottom of your window to open a fresh canvas (Sheet 4).

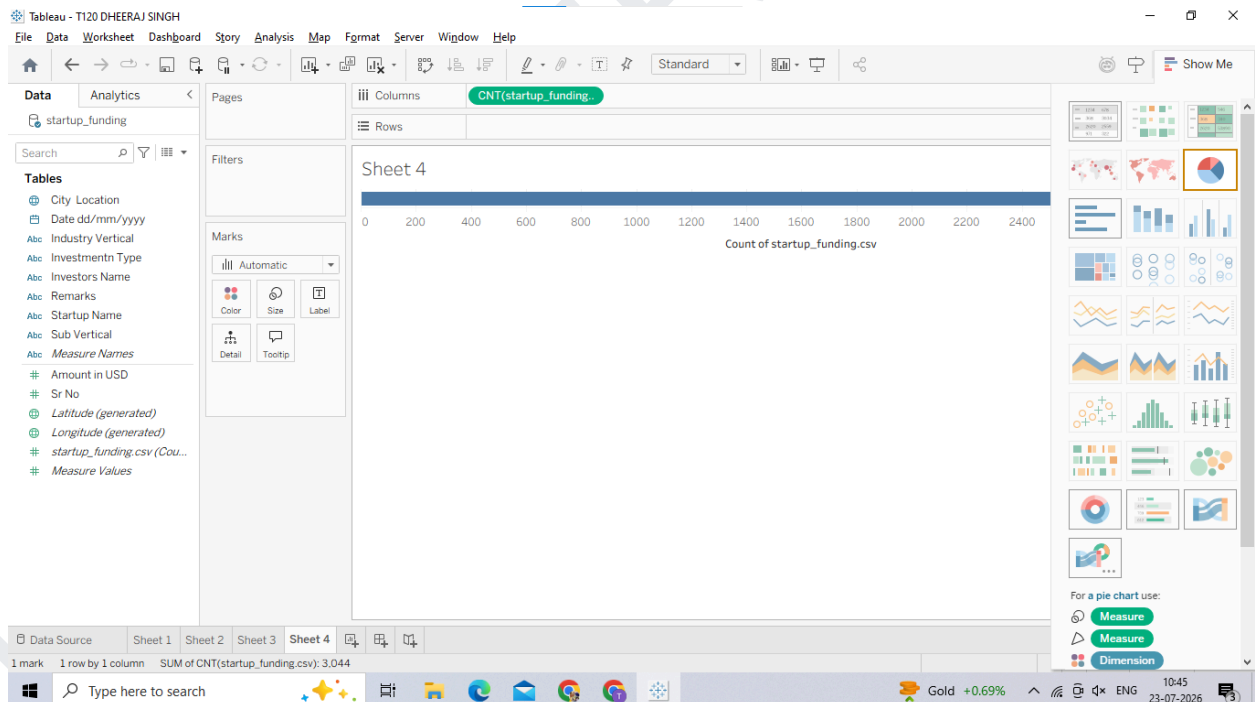


2. Look down to the bottom of your Data Pane list under the green measures section. Locate the automatic system measure field called startup\_funding.csv (Count) (in older versions of Tableau, this is called Number of Records).

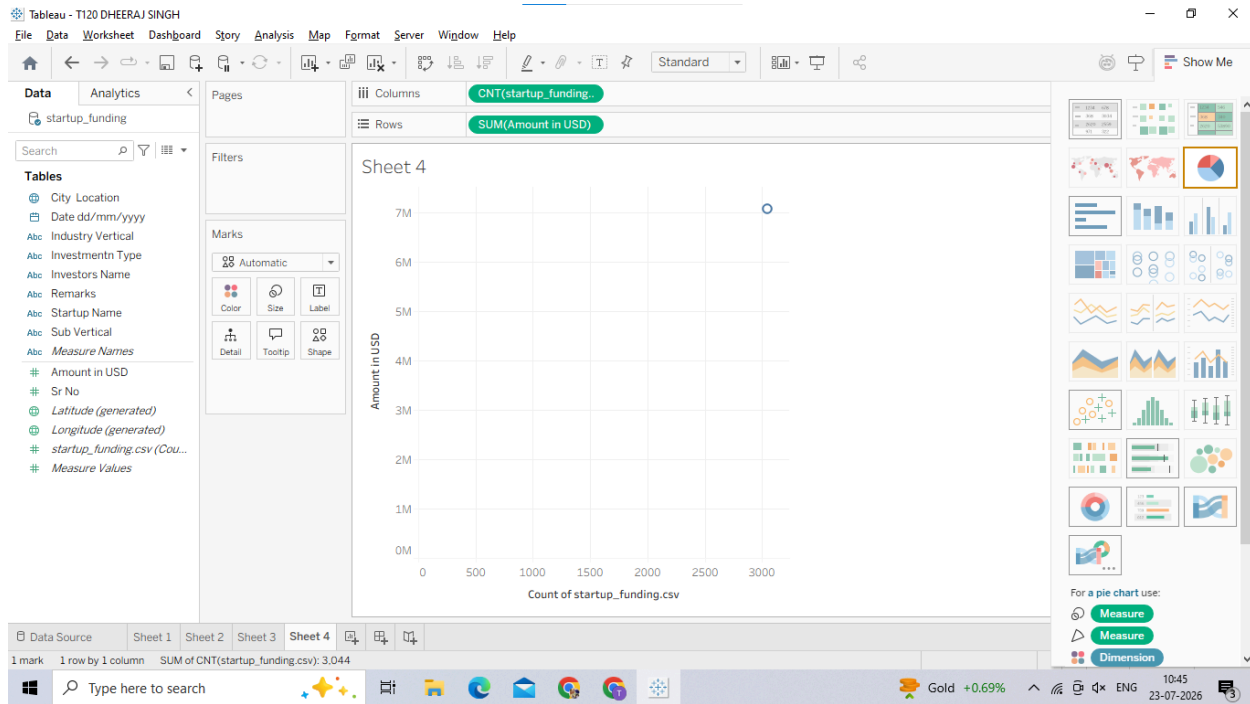
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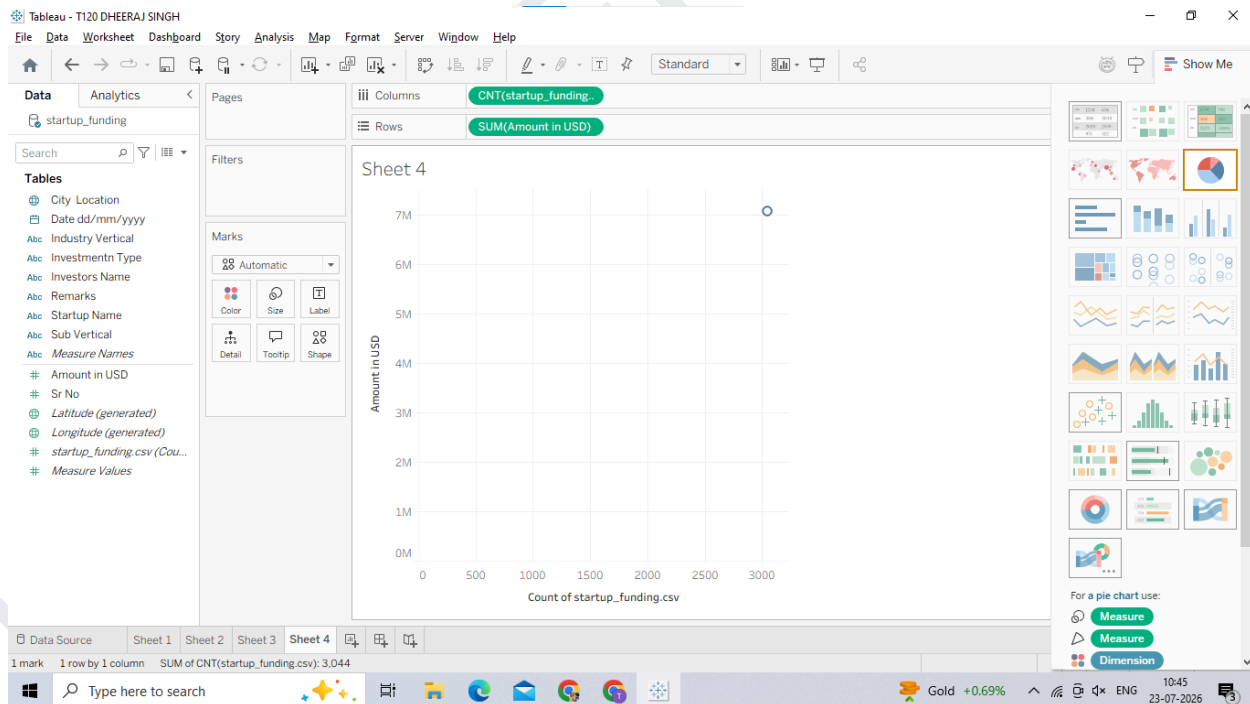
3. Drag that Count or Number of Records measure and drop it onto the Columns shelf.



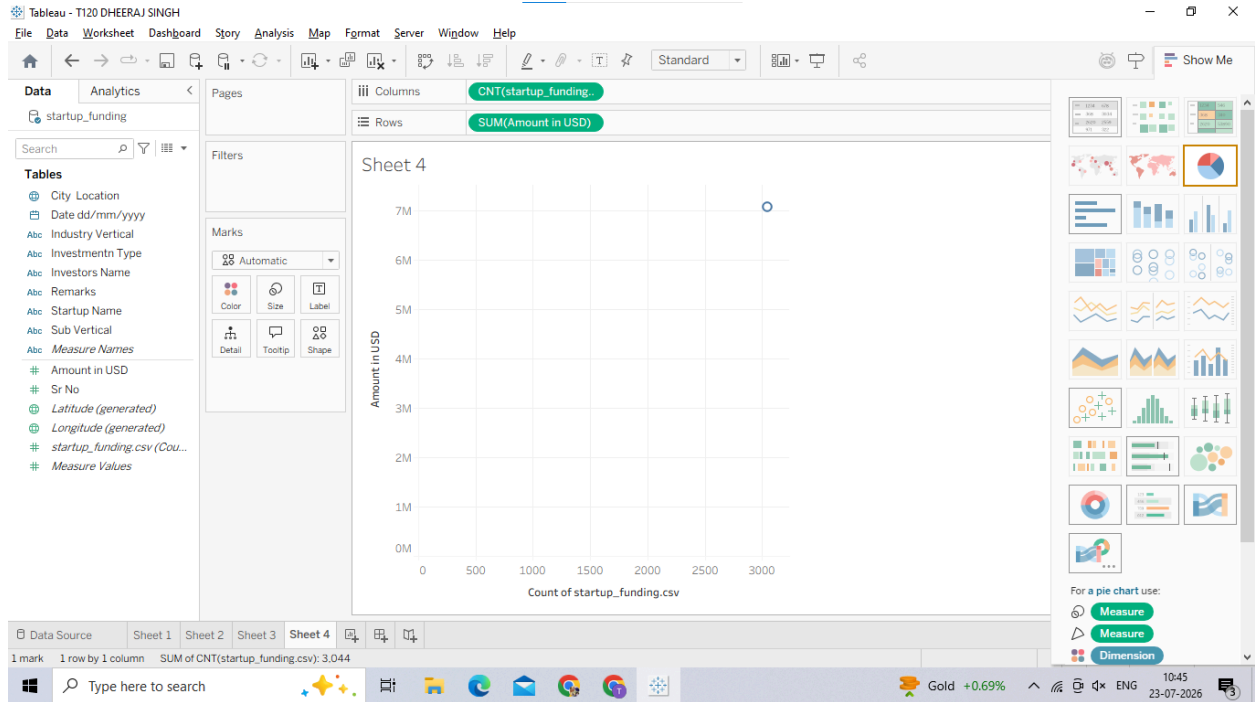
4. Drag your Amount in USD measure and drop it onto the Rows shelf.



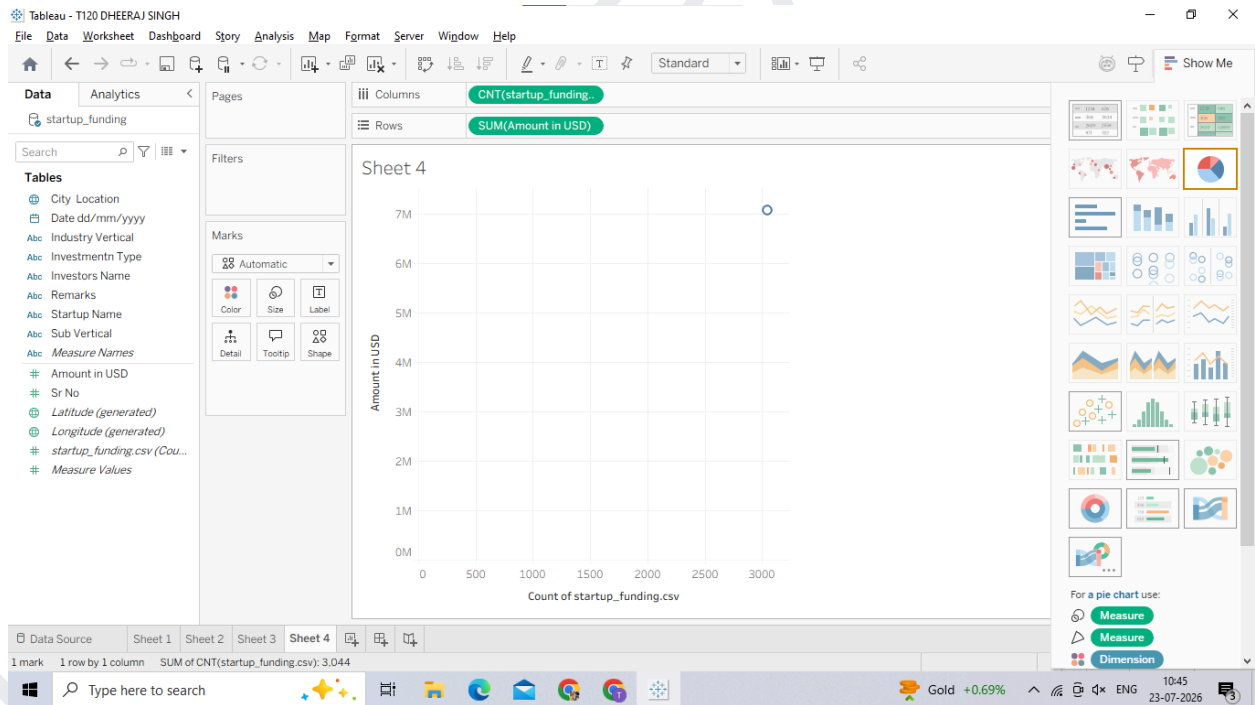
o Look closely: Right now, you will see a single, isolated data dot in the center of your chart canvas. This is because Tableau automatically sums up the values for the entire dataset into a single aggregated point.



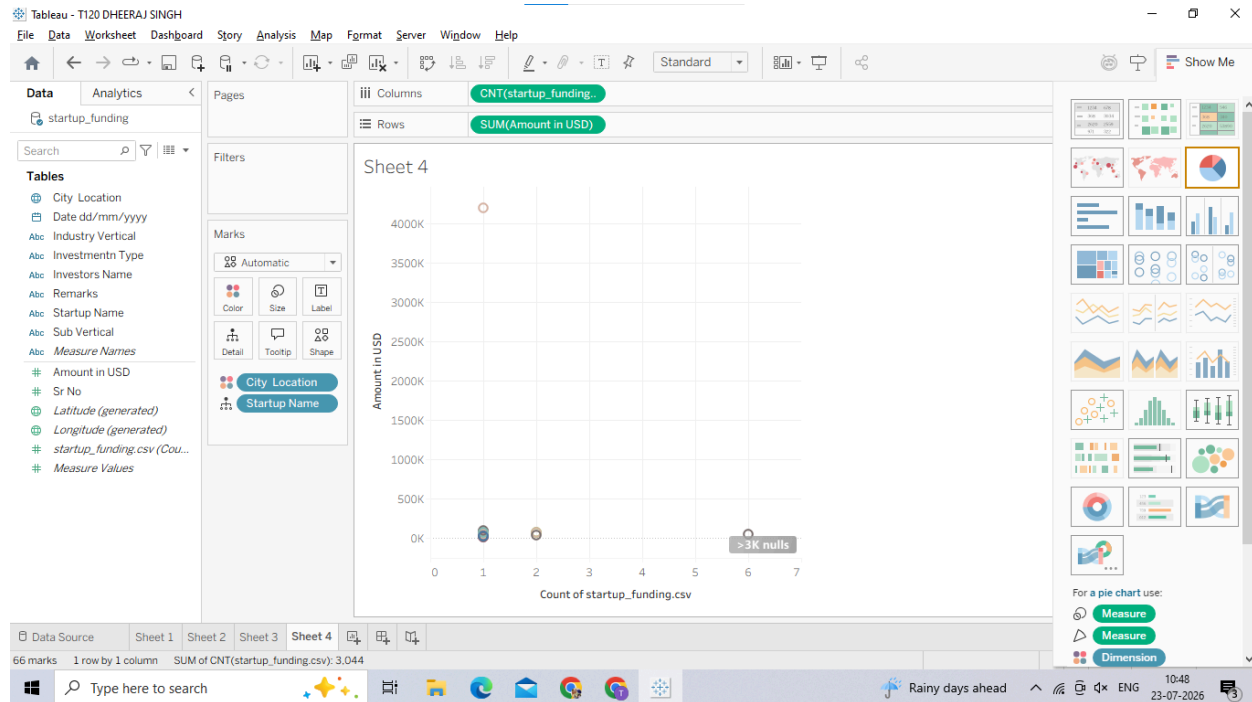
5. To break this single point down into individual startups, find the Startup Name dimension in the Data Pane.



6. Drag Startup Name and drop it directly onto the Detail tile within the Marks card.



7. Adding Context with Color: Take the City Location dimension from your Data Pane and drop it onto the Color tile in the Marks card.



## Why we are doing it

A Scatter Plot visualizes the distribution of data points across two intersecting numeric axes (X and Y). By dropping Startup Name onto the Detail tile, we instruct Tableau to disaggregate the total sum, plotting a distinct dot for every unique Indian startup. This allows us to visually inspect if there is a correlation between how many funding rounds a company secures versus the total cash volume they raise.

## Usage

Use a scatter plot when you want to discover trends, patterns, clusters, or extreme statistical outliers between two numeric variables (e.g., assessing if higher round counts directly correlate with higher valuation injections, or finding which startups are massive capital outliers).

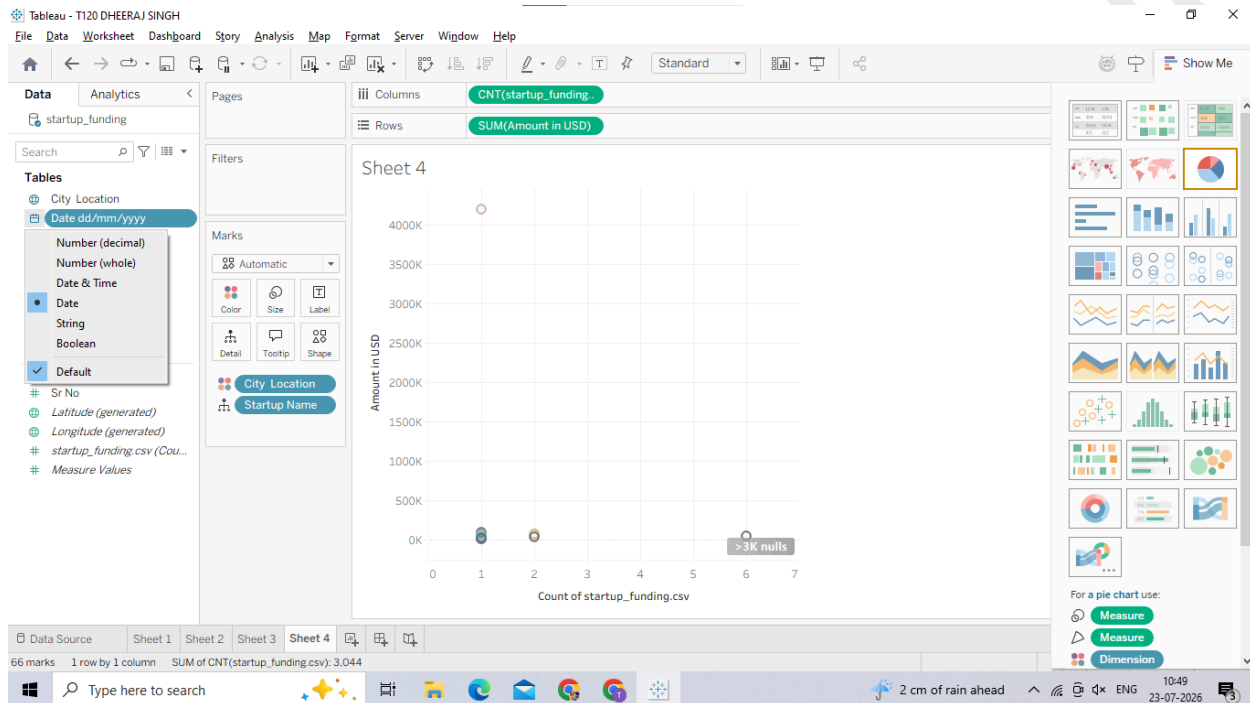
## Step D: Creating Dual-Axis and Combination Charts

### 1. Preparing the Fields

To build a dual-axis combination chart, we need a time element on our horizontal axis (\$X\$-axis) and two different metrics tracking along the vertical axis (\$Y\$-axis).

We will use the Date dd/mm/yyyy column for time. Before we begin, let's make sure Tableau treats this text field as a proper date:

1. Look at your left-hand Data Pane.
2. Find the field Date dd/mm/yyyy and click the small text icon (Abc) immediately to the left of its name.
3. Select Date from the dropdown menu to convert it into a structured time dimension.

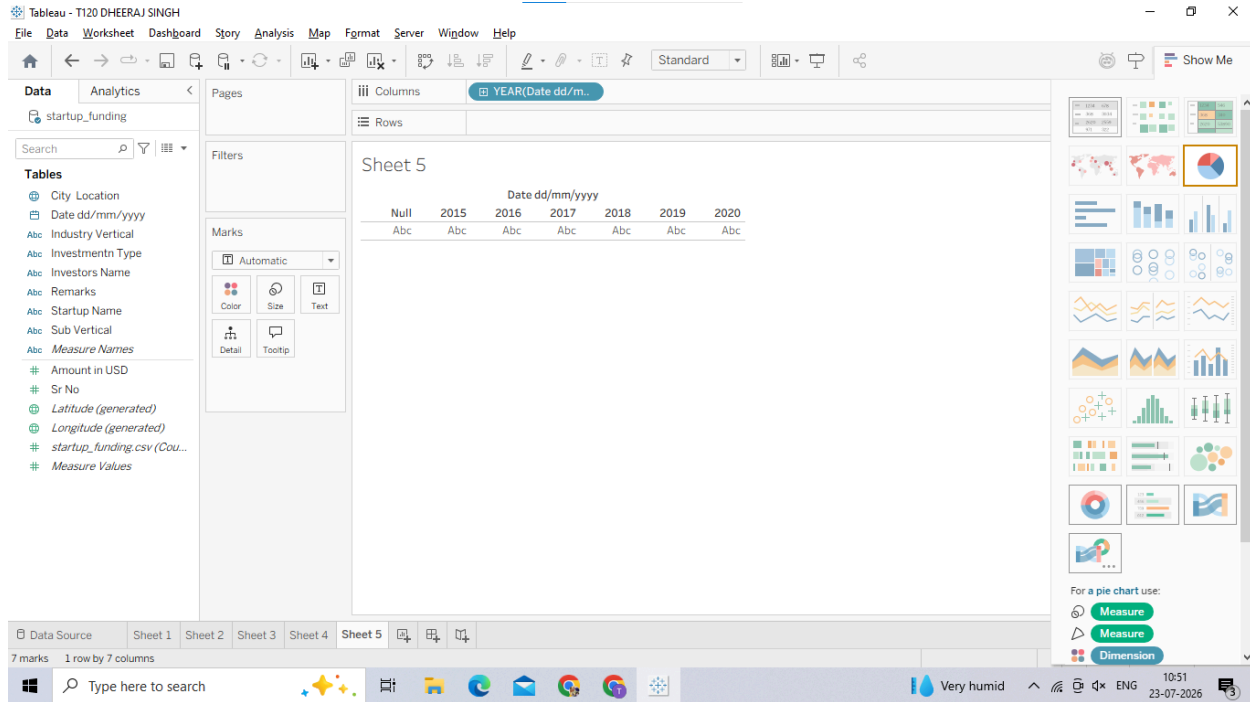


(In Some Cases the Date Option is selected by default as shown in the above given Snapshot)

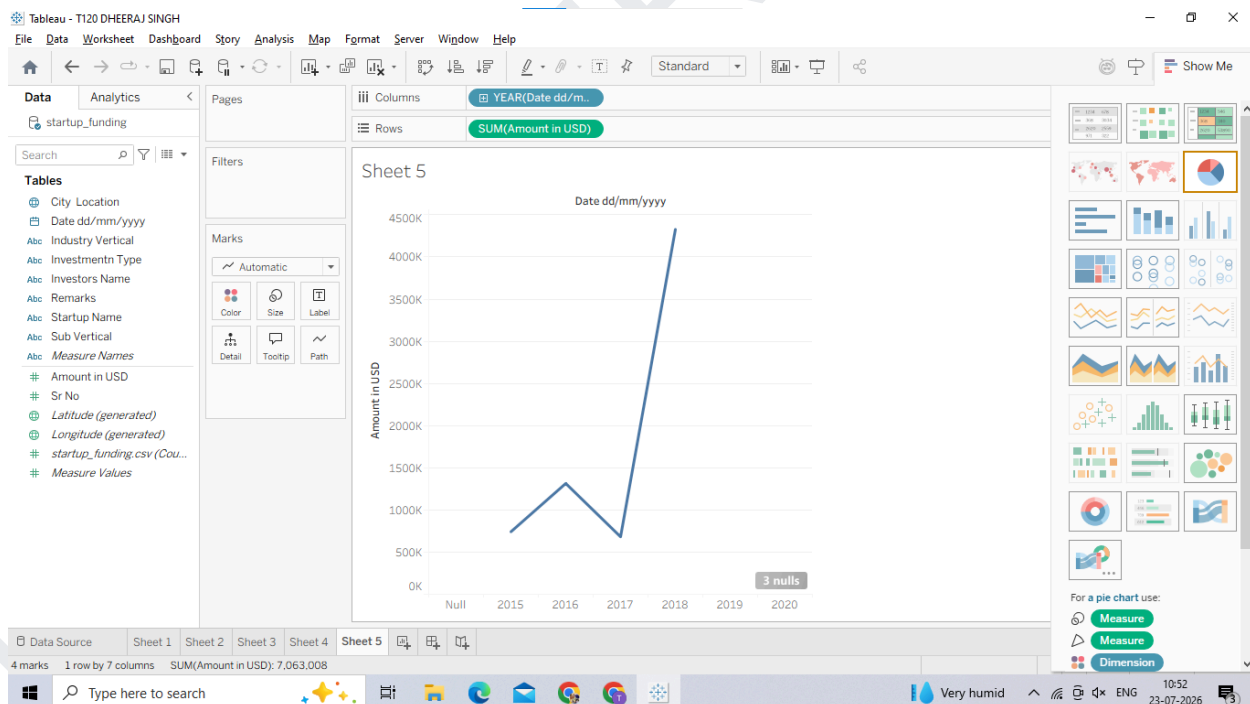
## 2. Building the Dual-Axis Chart

1. Click the New Worksheet icon at the bottom of the screen to open a fresh canvas (Sheet 5).
2. Drag your newly formatted Date dd/mm/yyyy dimension and drop it onto the Columns shelf. (Tableau will automatically group the dates into discrete Years).

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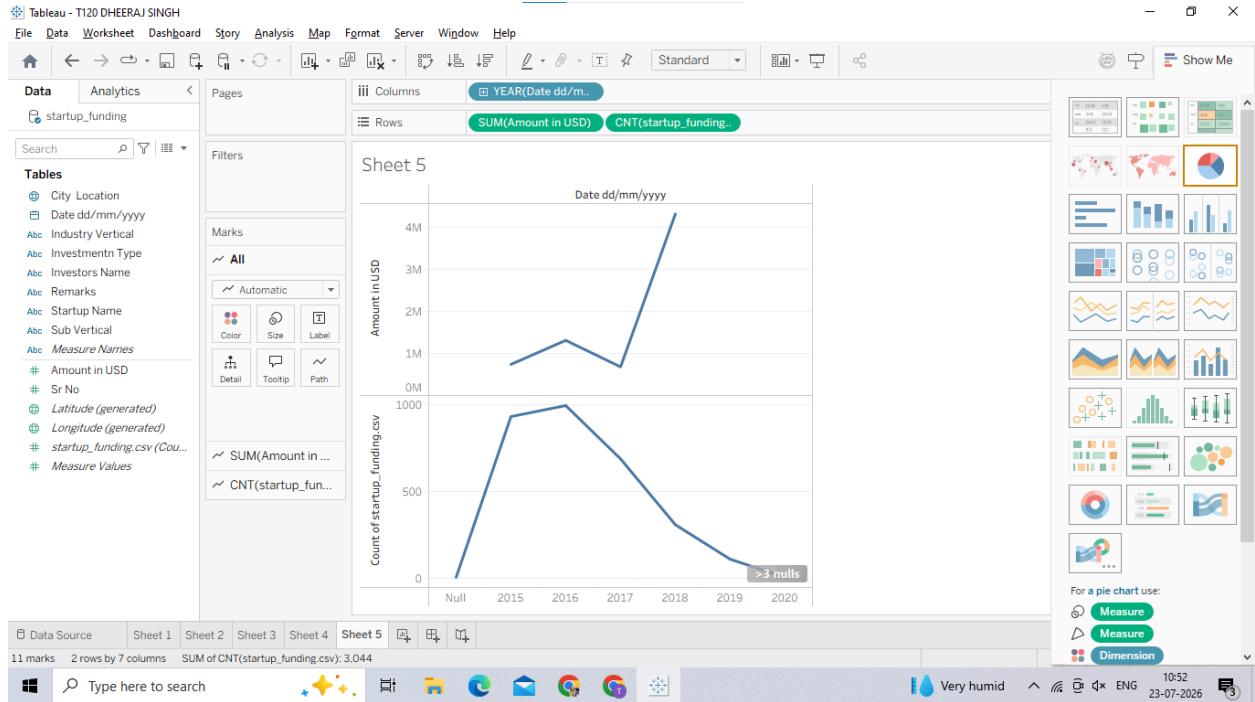


3. Drag the Amount in USD measure from the Data Pane and drop it onto the Rows shelf.

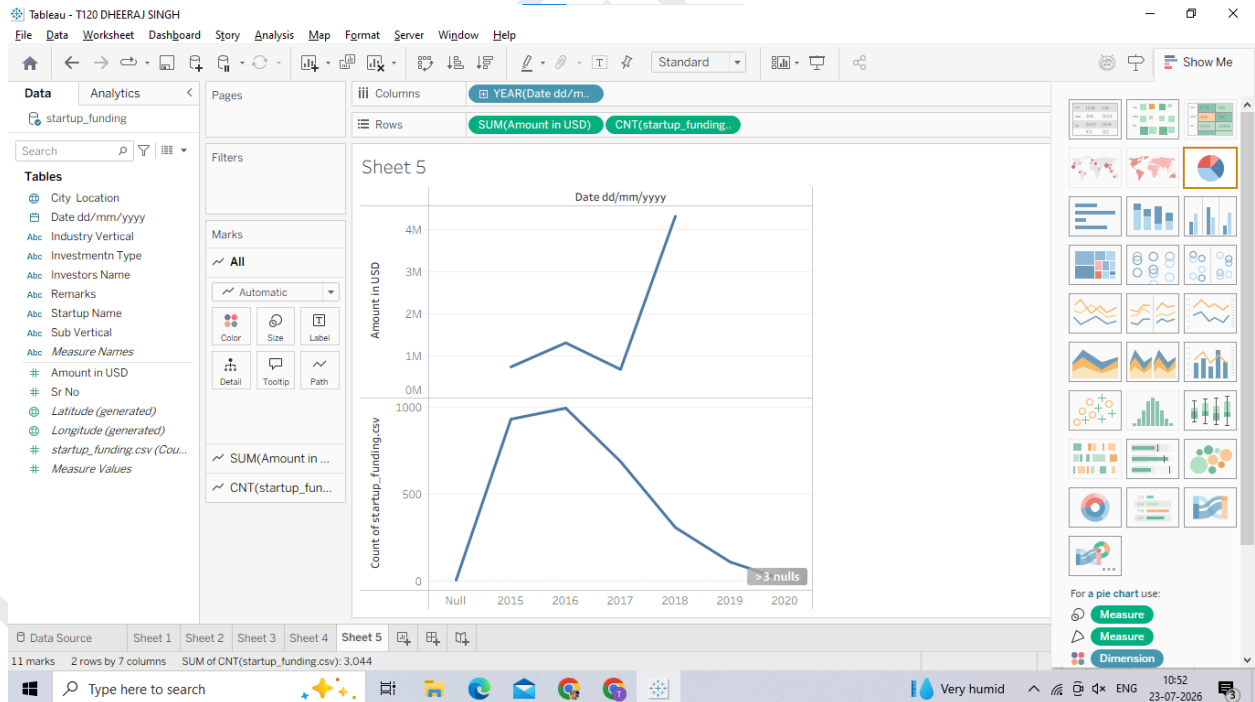


4. Now, take the automatic system measure startup\_funding.csv (Count) (or Number of Records) and drag it onto the Rows shelf, placing it directly to the right of the Amount in USD pill.

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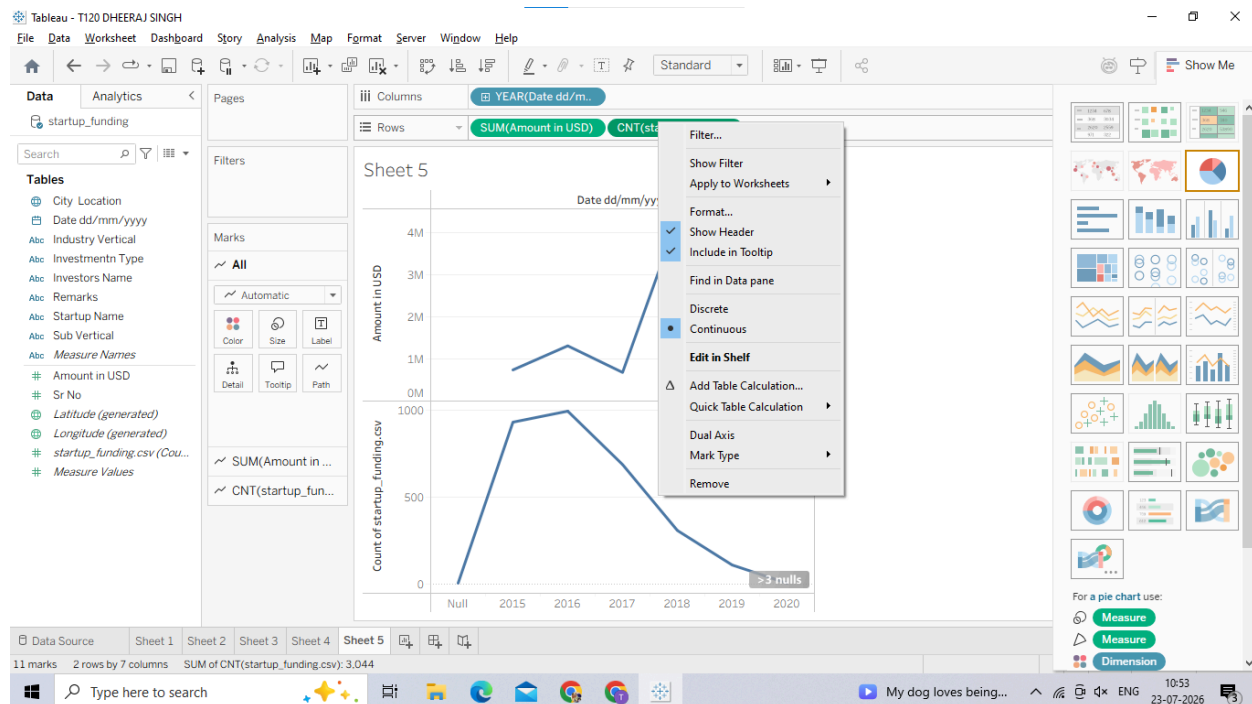


o Look closely: You will now see two separate line charts stacked vertically on top of each other.

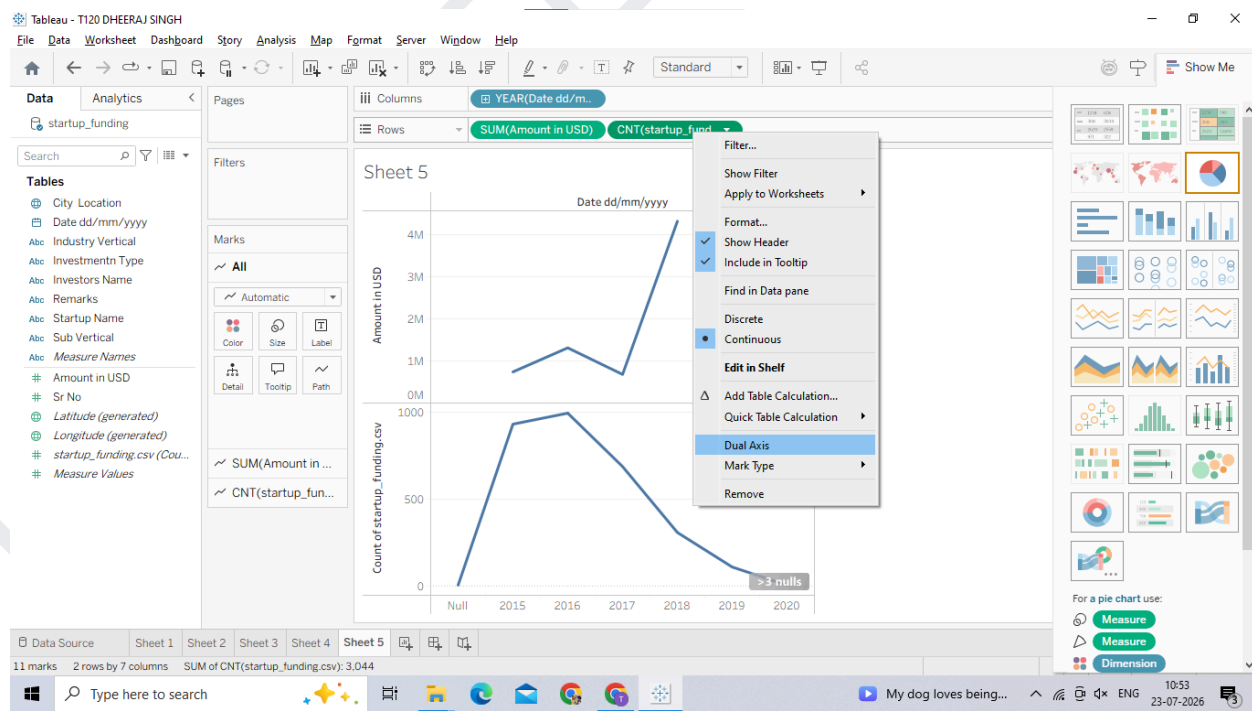


### 3. Combining and Formatting the Visuals

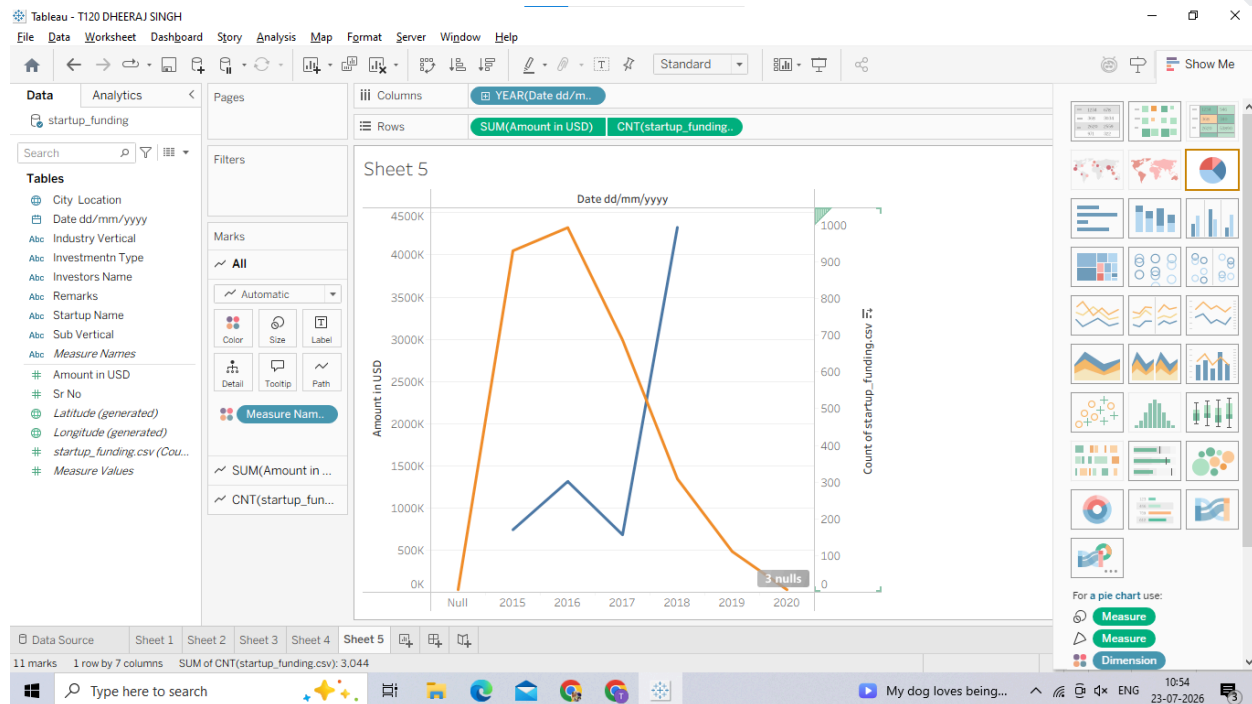
1. Right-click the second measure pill (startup\_funding.csv (Count)) sitting up on the Rows shelf.



2. Select Dual Axis from the context dropdown menu.



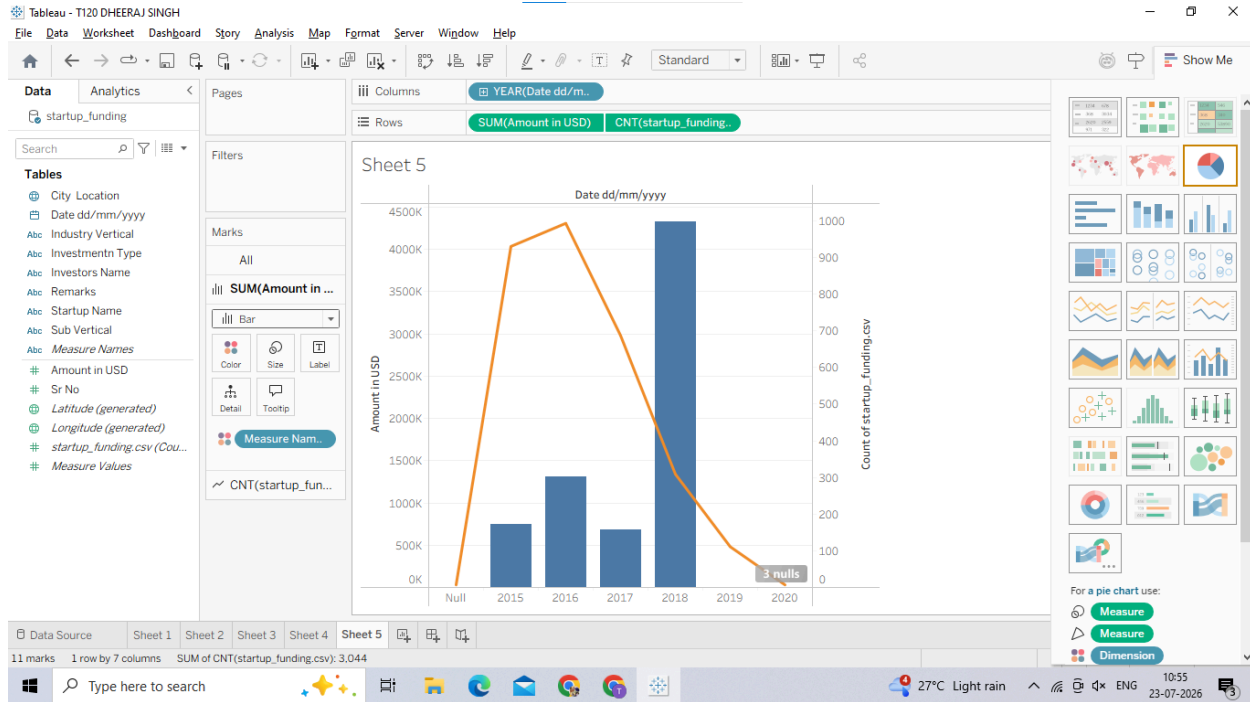
o Look closely: The two charts will instantly merge into a single canvas area, adding a second vertical axis on the far-right side of your screen.



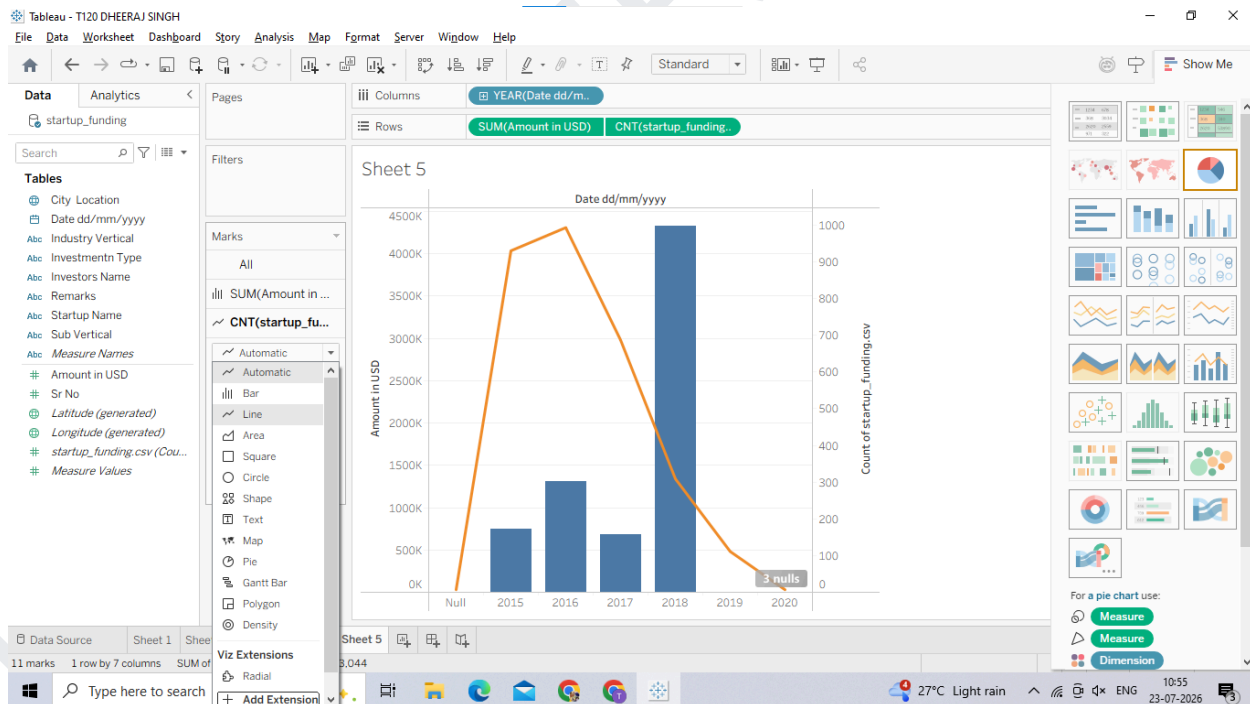
3. Look at your Marks Card area. Notice that it has split into three distinct tabs: All, SUM(Amount in USD), and CNT(startup\_funding.csv).

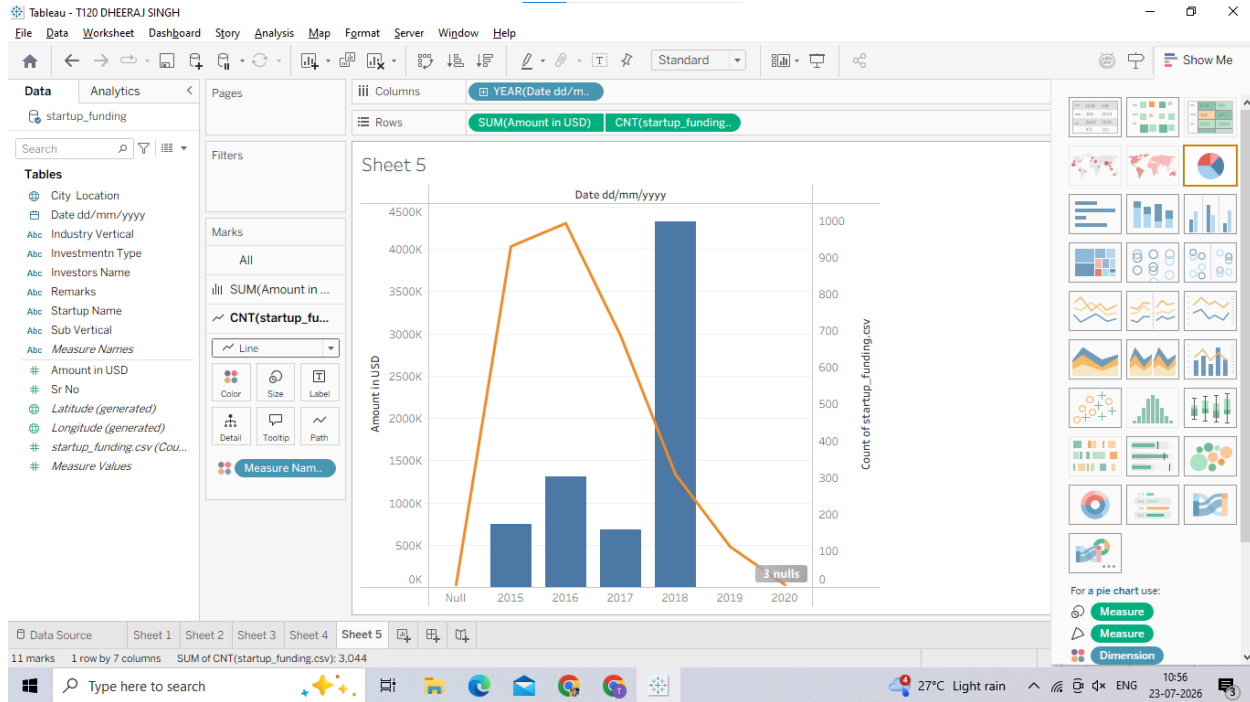


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5. Next, click directly on the CNT(startup\_funding.csv) tab in the Marks card. Change its visual type dropdown selection from Automatic to Line.





## Why we are doing it

A Dual-Axis Combination Chart plots two completely different metrics with unique units of scale (e.g., massive dollar amounts in millions on the left vs. small, double-digit transaction counts on the right) across a shared timeline variable.

## Usage

Use a dual-axis combination chart when you want to look for patterns or structural shifts between volume and scale simultaneously. For instance, this specific layout lets you immediately see if an increase in the total funding amount in a given year was driven by a high volume of small startup deals, or if a few massive investments bloated the total capital while overall deal counts dropped.

## Step E: Developing Histograms for Distribution Analysis

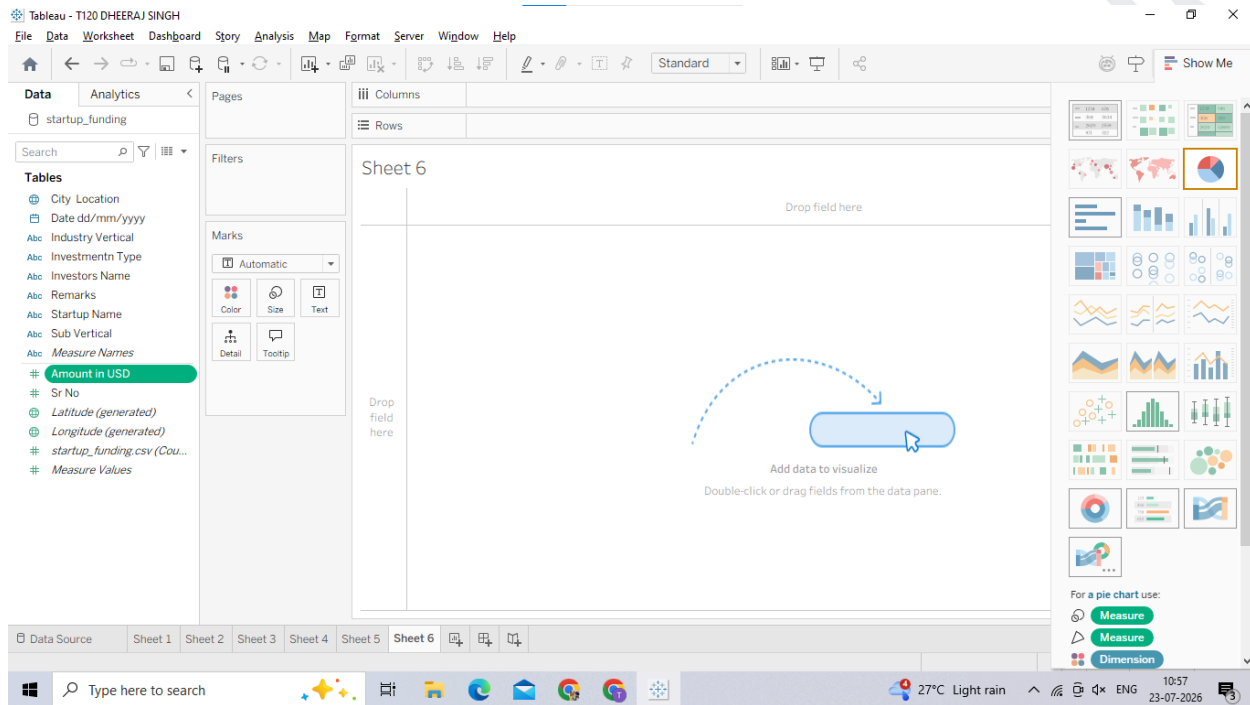
### 1. Understanding Histograms in Tableau

A histogram splits a single continuous numerical measure into equal, consecutive intervals called Bins (e.g., separating funding amounts into bins of 0–10M, 10M–20M, etc.) and displays the frequency count of records falling within each bucket as a bar chart.

Because a histogram relies on a single continuous field to build its own bins, we don't manually drop fields onto both shelves like a normal bar chart.

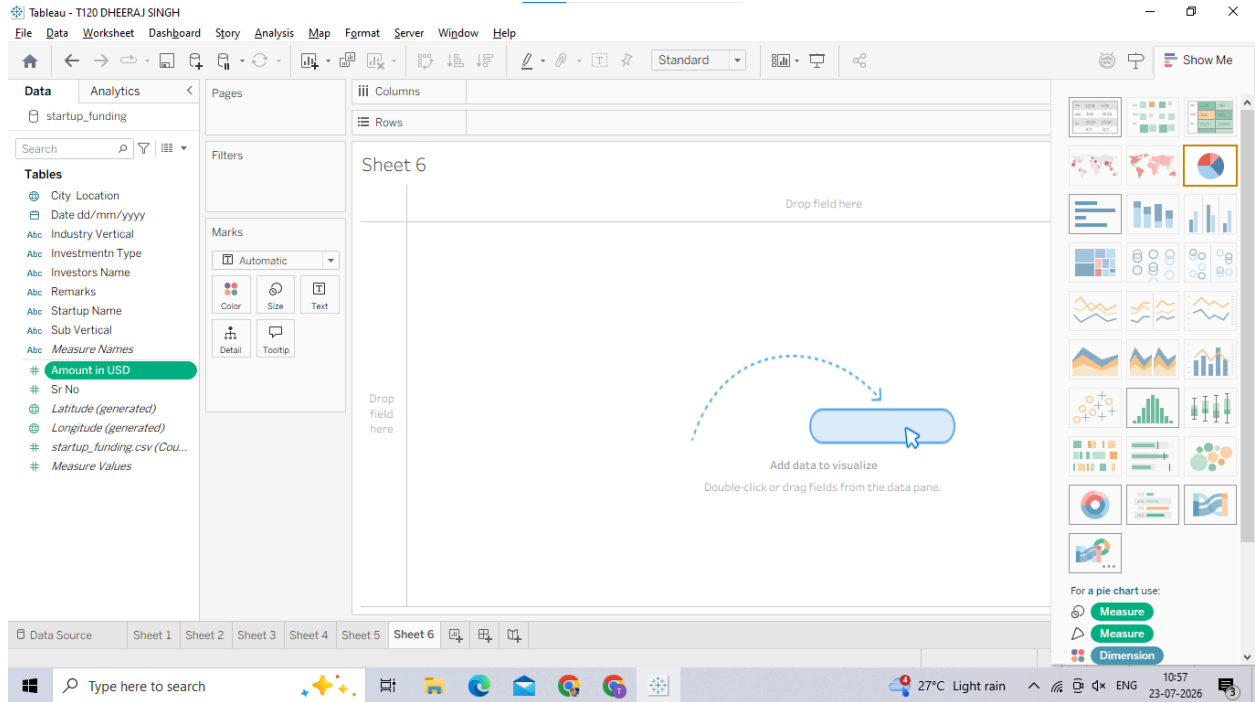
## 2. Building the Histogram

1. Click the New Worksheet icon at the bottom of the screen to open a fresh canvas (Sheet 6).
2. Go to the Data Pane and click once directly on the Amount in USD measure to highlight it.

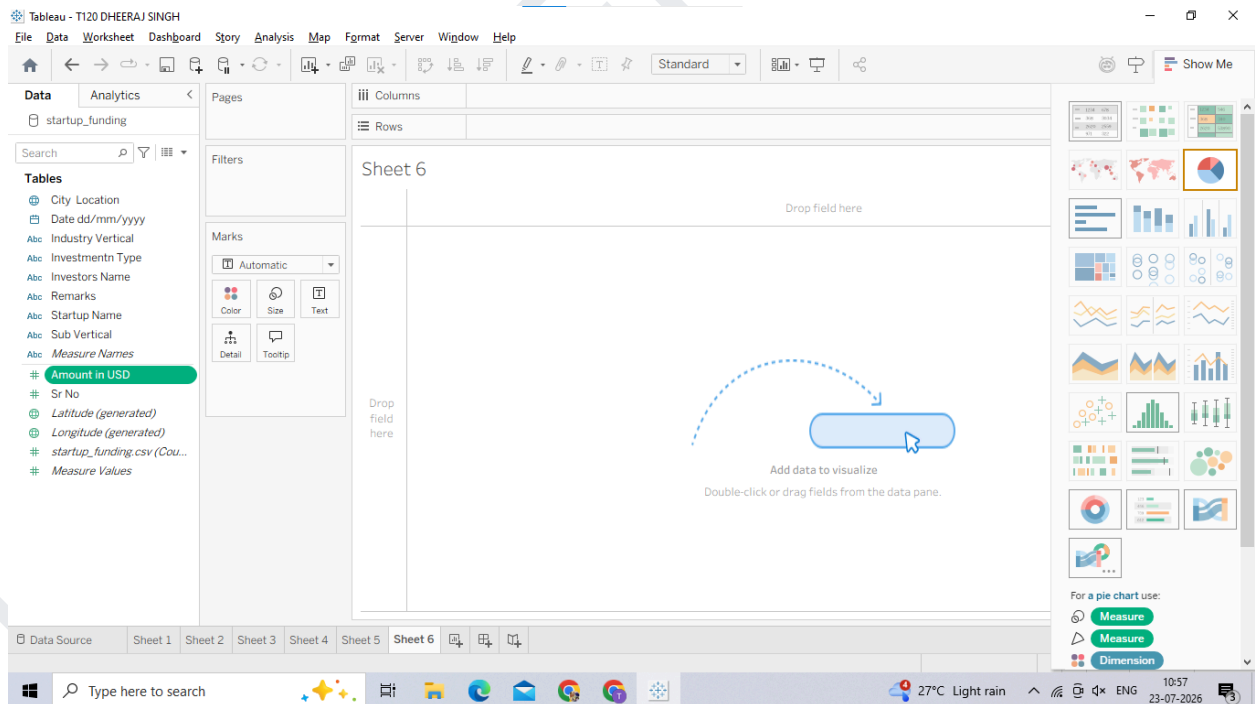


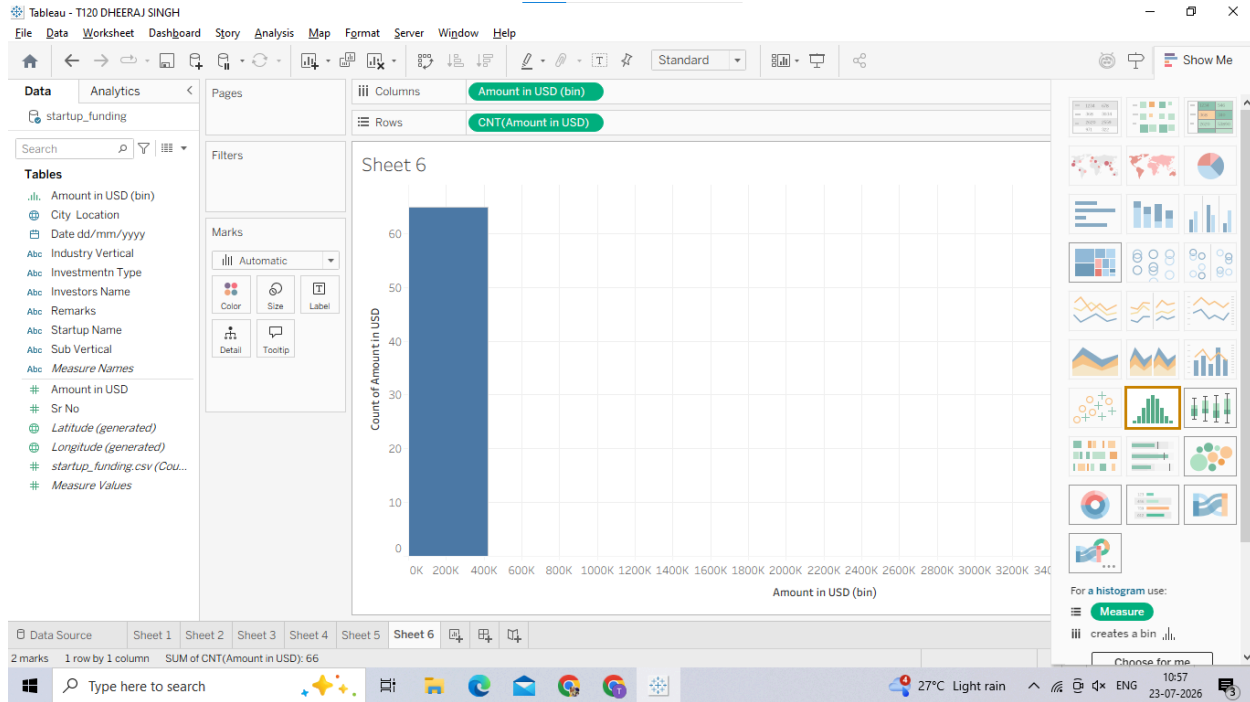
3. Move your mouse to the top-right corner of the window and click the Show Me button to expand the chart library panel.(Which is already Expanded)

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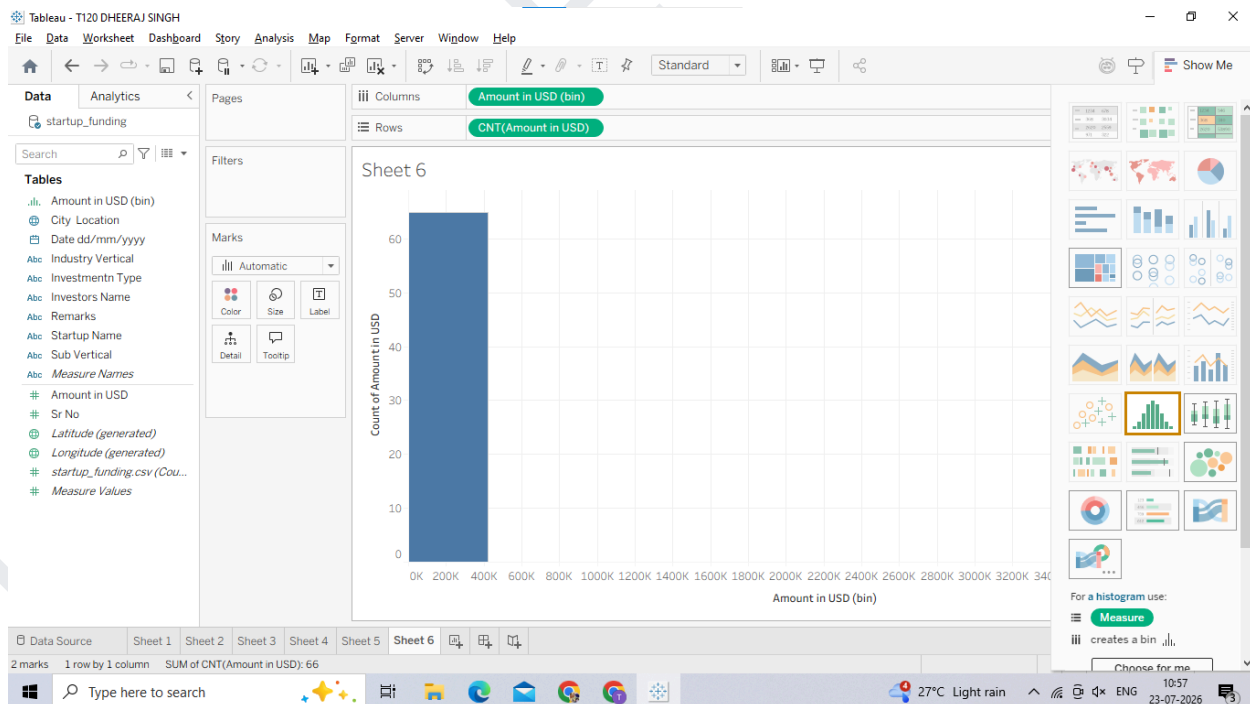


4. From the options, locate and click the Histogram icon (found in the middle row, displaying vertical green bars).



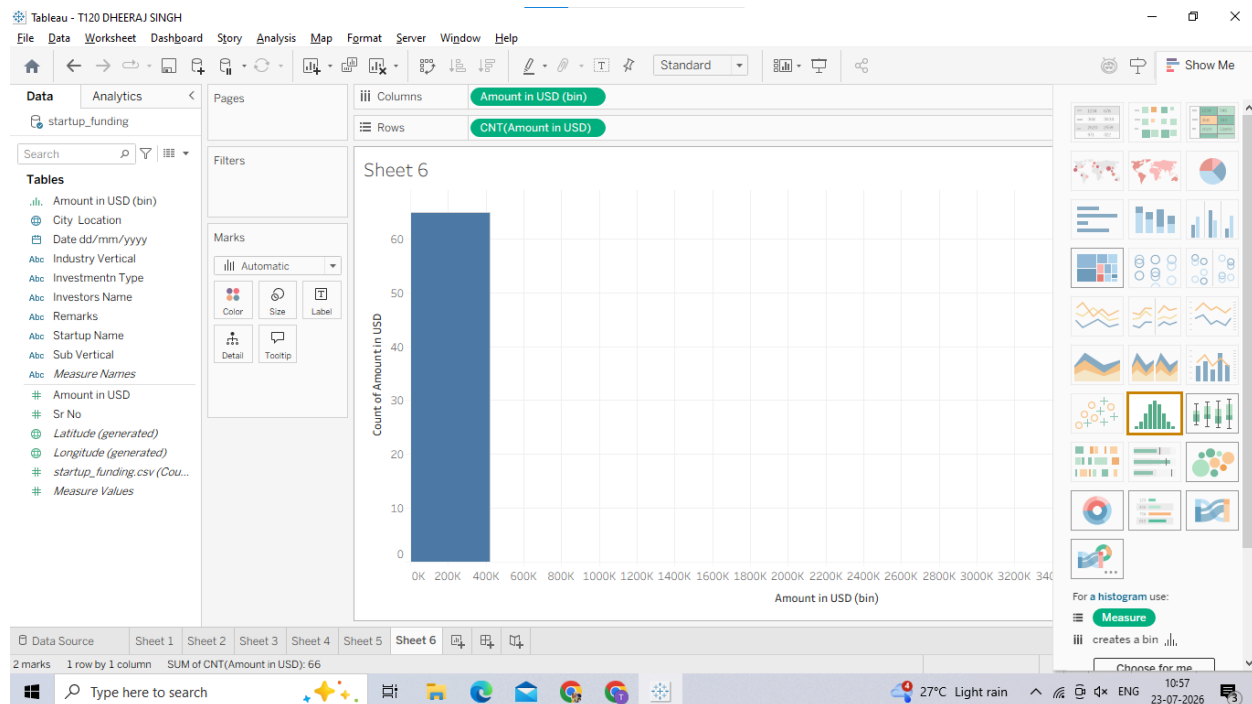


5. Look back at your Data Pane. Notice that Tableau has automatically generated a brand-new field under your dimensions list called Amount in USD (bin) and placed it onto your Columns shelf.



6. Look at the Rows shelf. Tableau has automatically placed the aggregated measure

CNT(Amount in USD) there to track record counts.



## Why we are doing it

A Histogram allows us to visually inspect the underlying statistical distribution shape of a numeric variable. By converting a continuous field into discrete intervals, we can quickly discover if our dataset has a normal distribution curve, if it is skewed heavily in one direction, or where the vast majority of our transaction frequencies cluster.

## Usage

Use a histogram when you want to conduct density or frequency analysis on raw volumes. For instance, in our Indian Startup Funding data, the histogram instantly shows us whether the startup ecosystem is dominated by a massive volume of micro seed-stage funding rounds, or if large multi-million-dollar mega-deals make up the bulk of structural market activity.

## Step F: Building Box-and-Whisker Plots

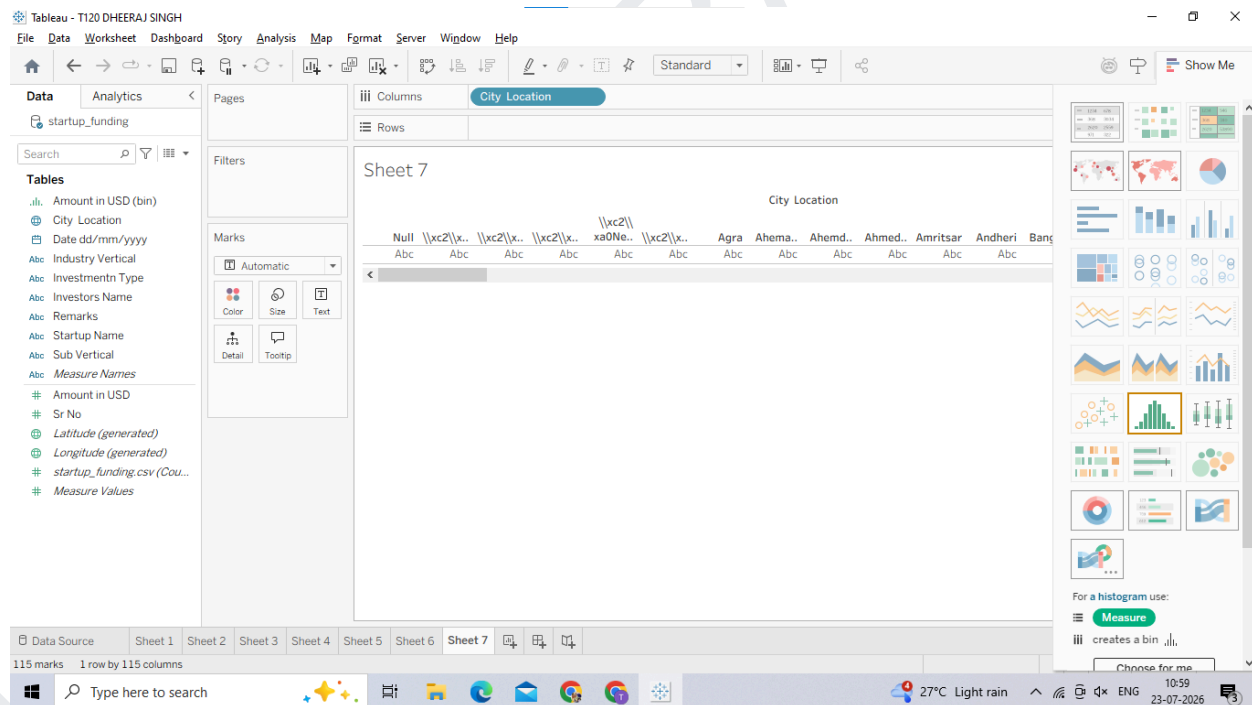
### 1. Cleaning the View for Box Plots

A Box-and-Whisker plot is a statistical chart used to show the distribution of data values based on a five-number summary (Minimum, First Quartile, Median, Third Quartile, and Maximum).

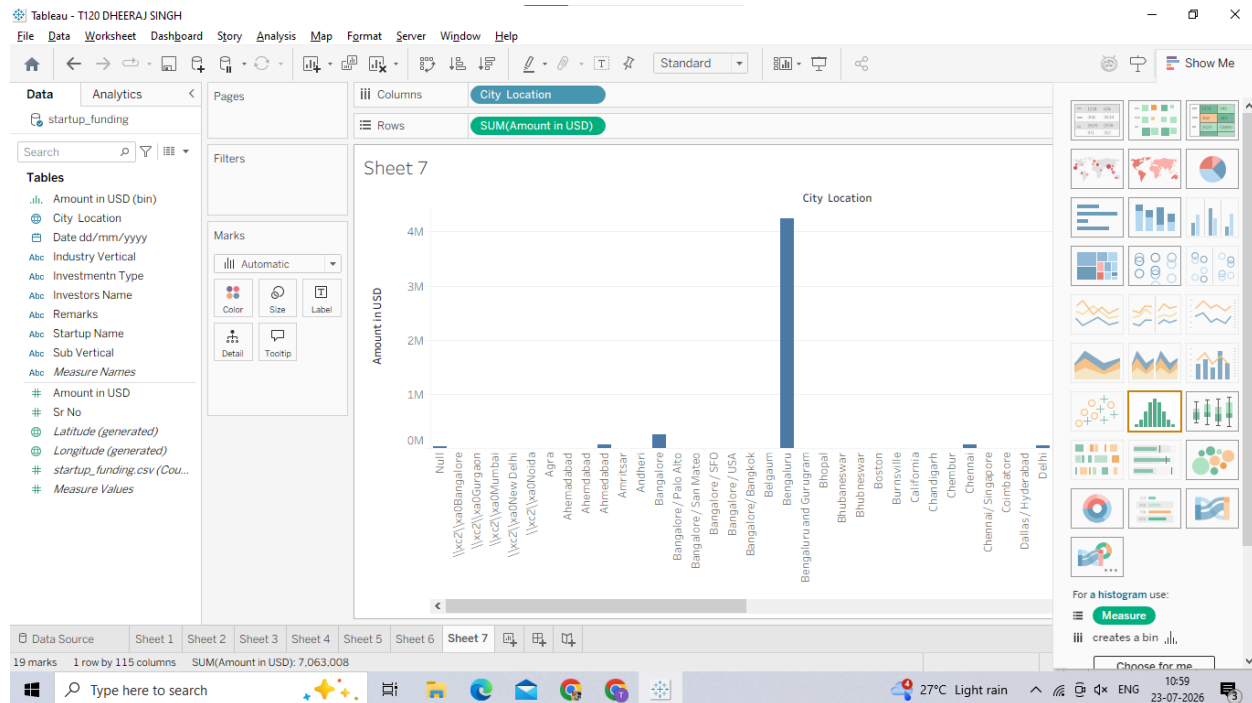
Because box plots measure structural distribution across data points, we must tell Tableau to look at the individual funding records rather than squishing everything into a single consolidated sum bar.

### 2. Building the Box-and-Whisker Plot

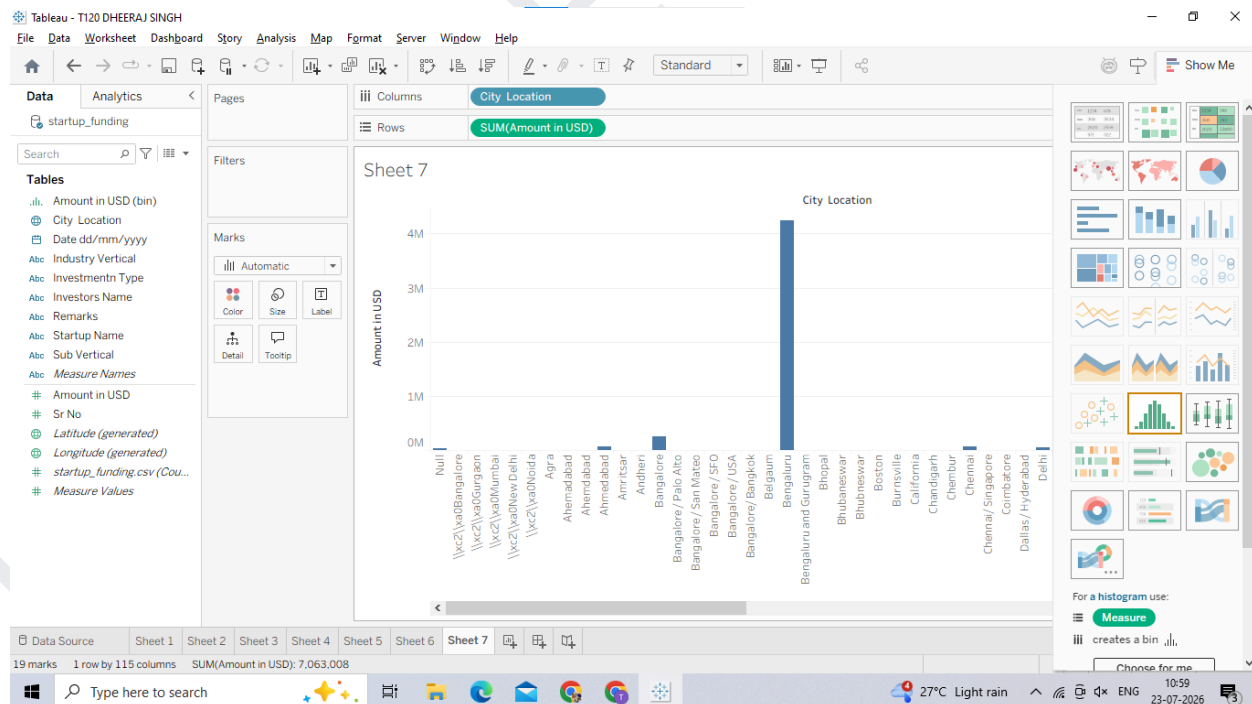
1. Click the New Worksheet icon at the bottom of the screen to open a fresh canvas (Sheet 7).
2. Drag the City Location dimension from the Data Pane and drop it onto the Columns shelf.



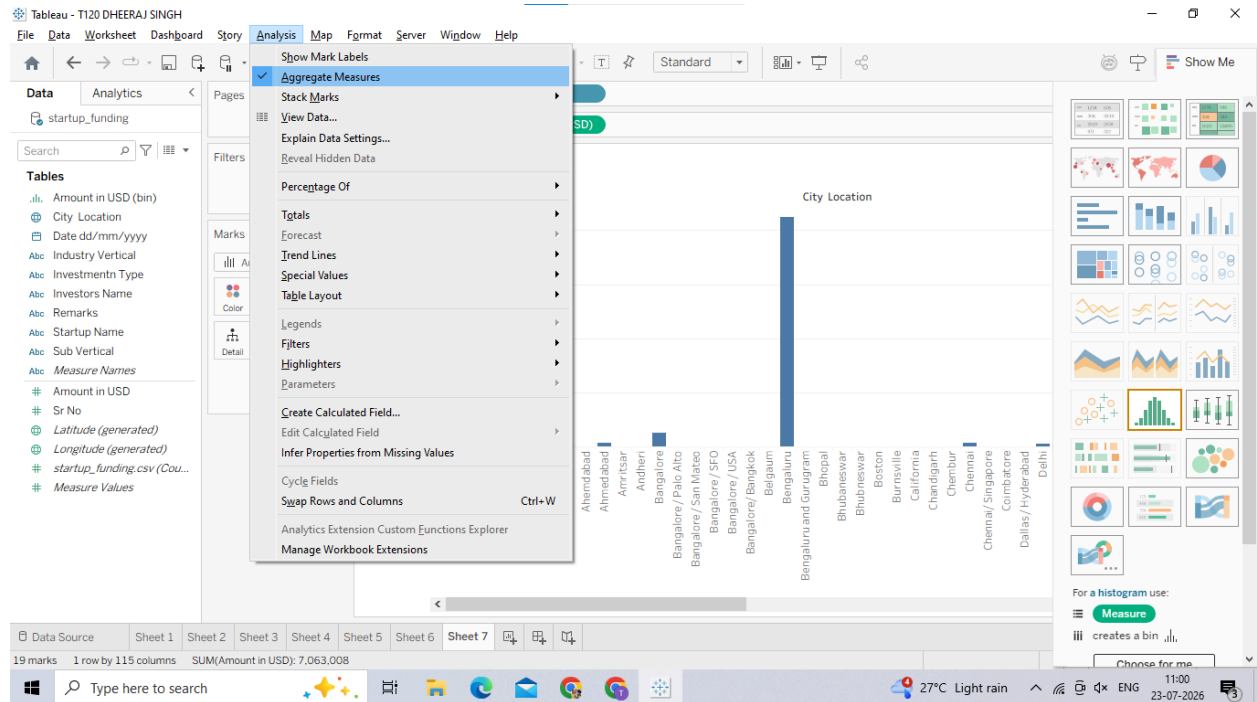
### 3. Drag the Amount in USD measure and drop it onto the Rows shelf.



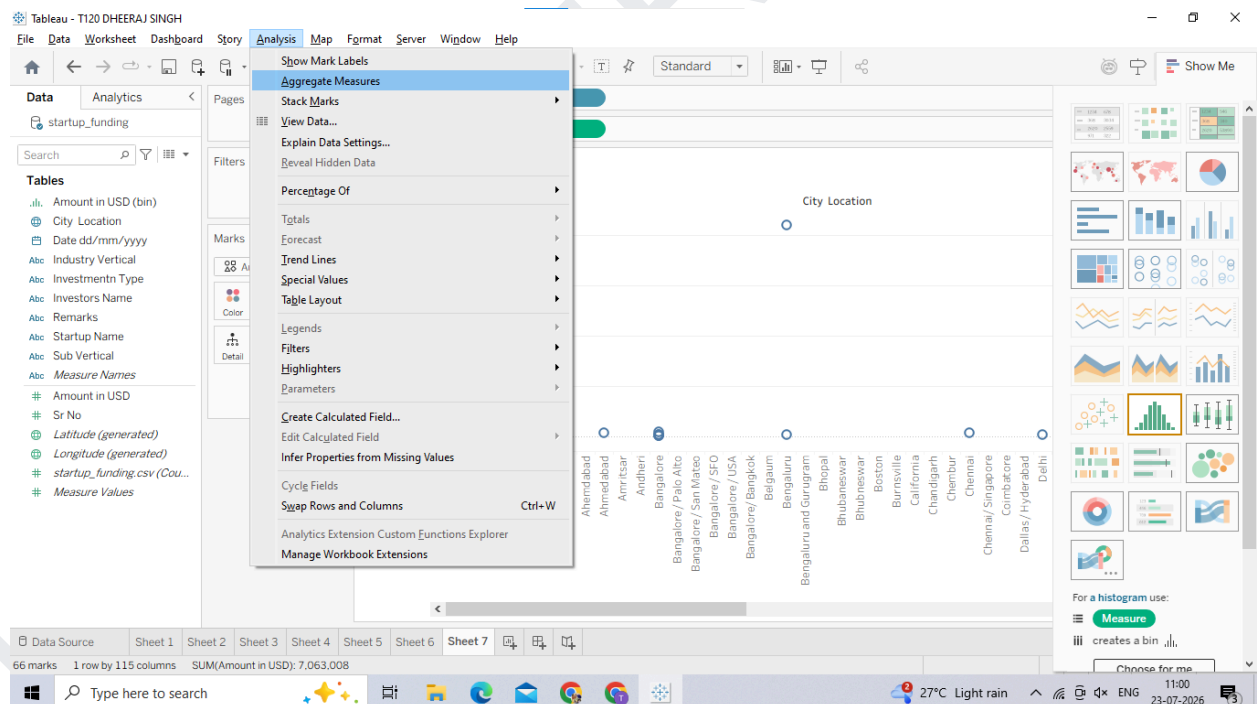
### 4. Disaggregate the Data Points: Go to the top menu bar click Analysis, and uncheck Aggregate Measures.



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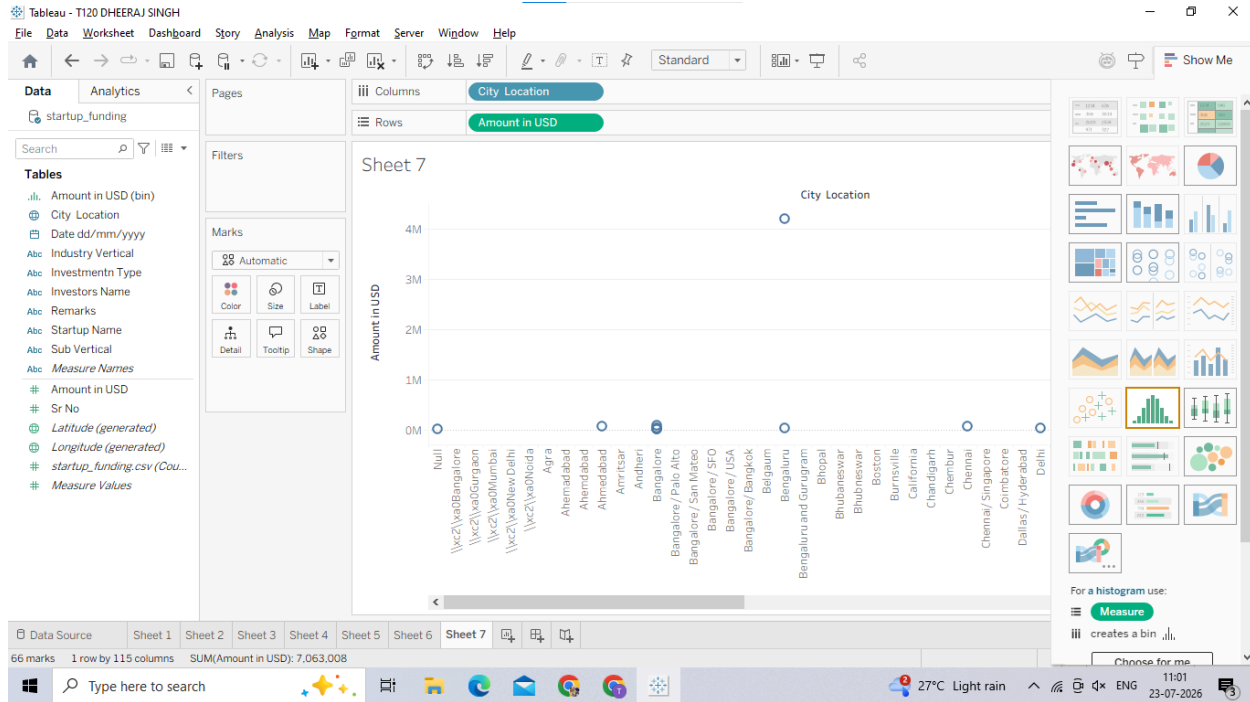


After Unchecking

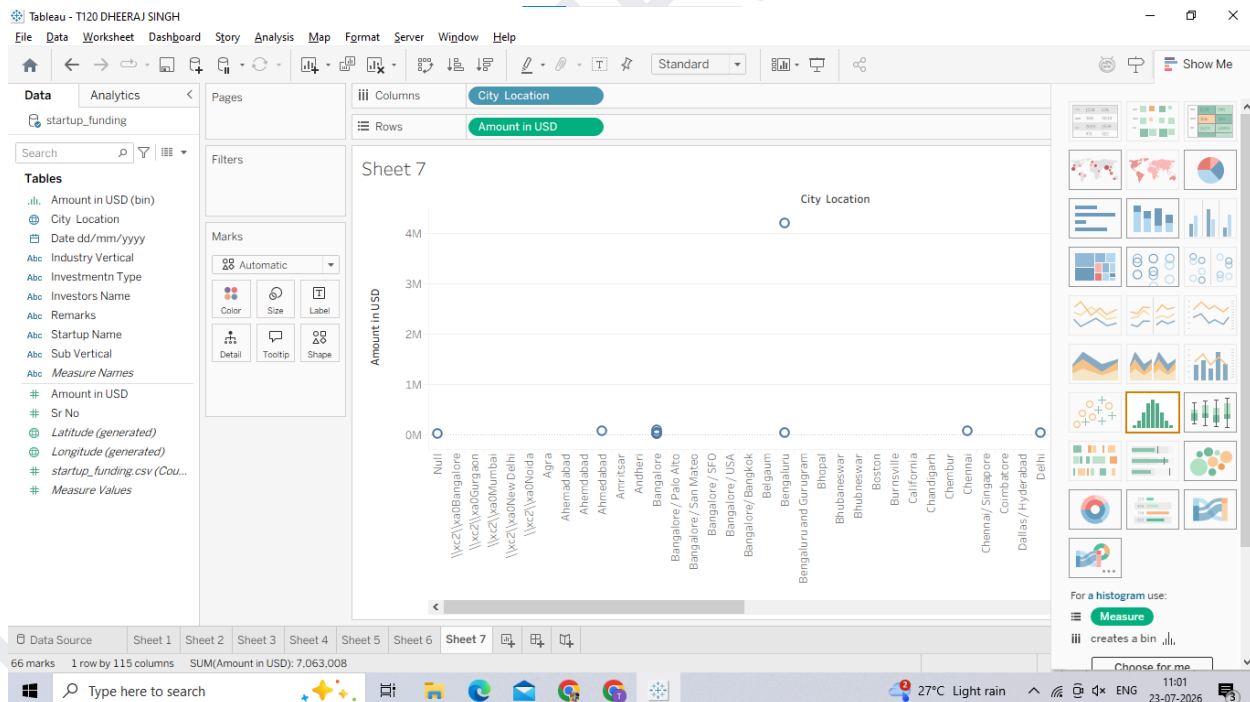


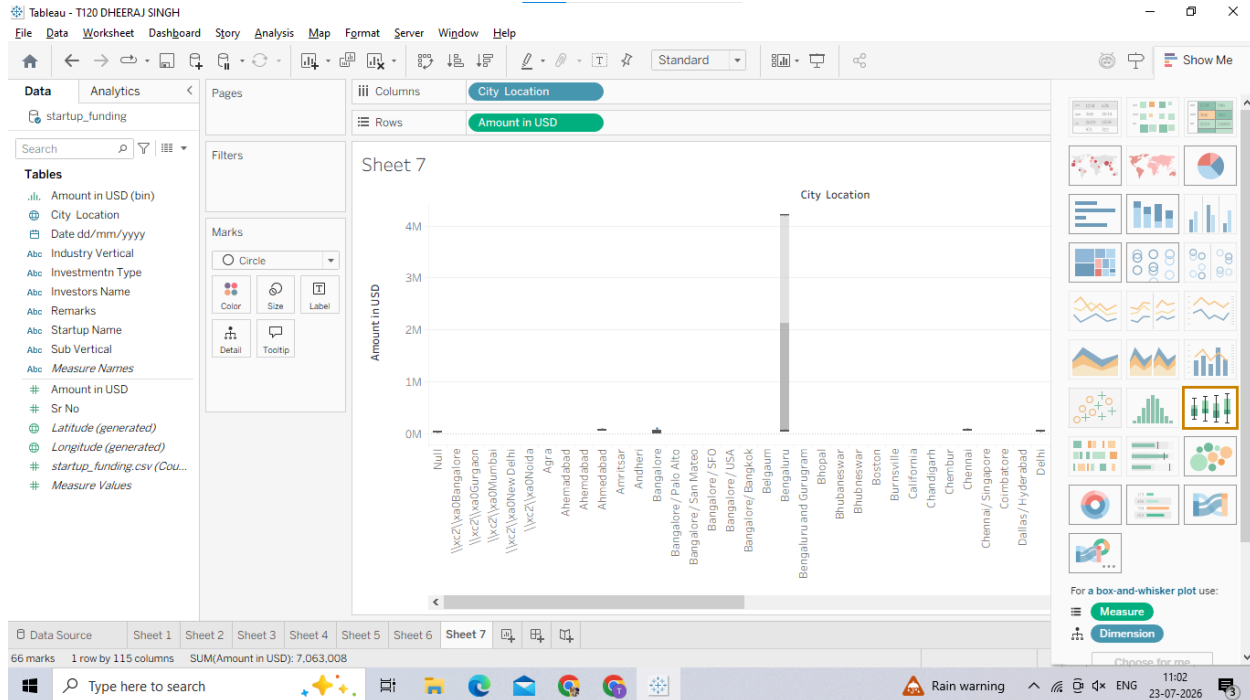
o Look closely: Your layout will instantly transform from a single compressed mark into a vertical scattered line of hundreds of individual circular marks for each city column.

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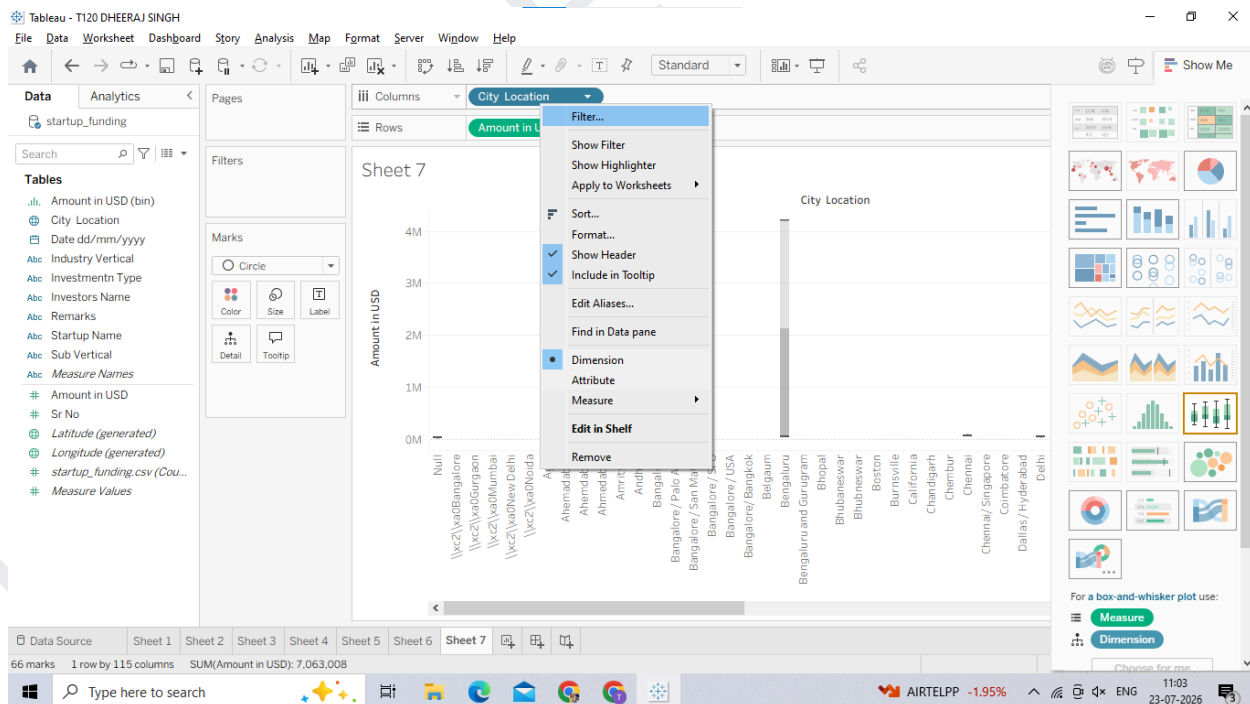


5. Add the Box Plot: Look at the top-right corner of the window, click the Show Me button, and select the Box-and-Whisker Plots icon (found in the middle row, showing a box shape with extended antenna lines).

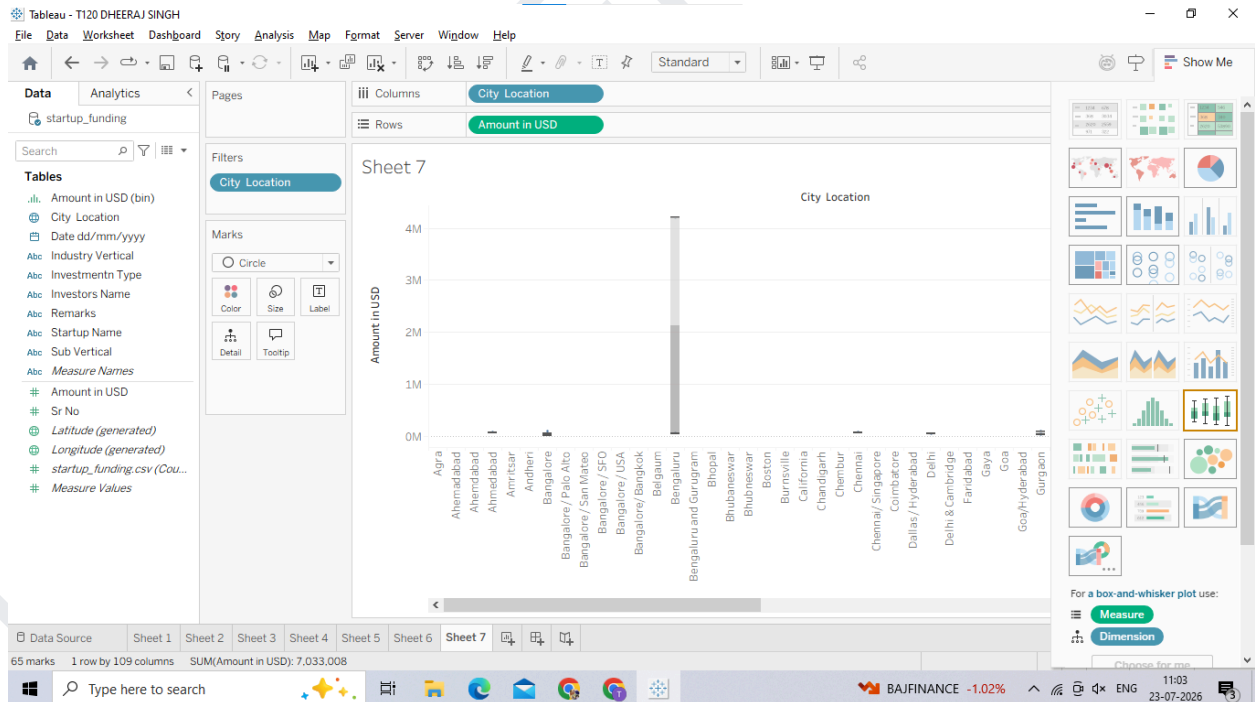
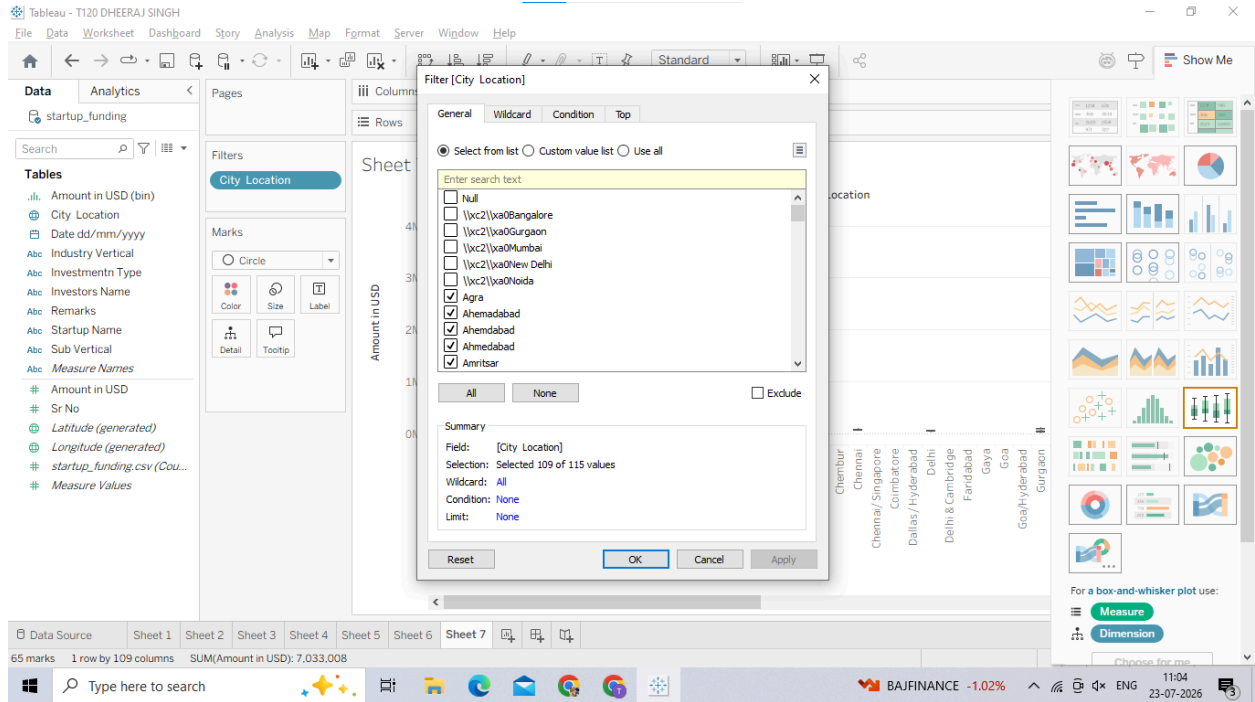




6. Refining the Focus: Since there are many small towns with only one funding record, right-click the City Location pill on your Columns shelf, select Filter, choose only the top tier hubs (e.g., Bengaluru, Mumbai, New Delhi, Gurgaon), and click OK to clean up your distribution view.



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## Why we are doing it

A Box-and-Whisker Plot visually clusters the core concentration of data values. The "box" highlights where the middle 50% of your values lie (the Interquartile Range), the middle line marks the true median, and the "whiskers" extend out to mark the standard data spread boundary. Any individual points floating completely outside those whiskers are flagged automatically as extreme statistical anomalies.

## Usage

Use a box-and-whisker plot when you want to compare the spread, density, and skewness of numeric data across different operational categories. For example, in our Indian Startup ecosystem, a box plot lets you compare Mumbai versus Bengaluru directly—showing you at a glance which city has a higher median deal value, which hub has more predictable funding ranges, and which market contains extreme, multi-million dollar venture capital outliers.

## Step G: Using Gantt Charts and Advanced Date Visualizations

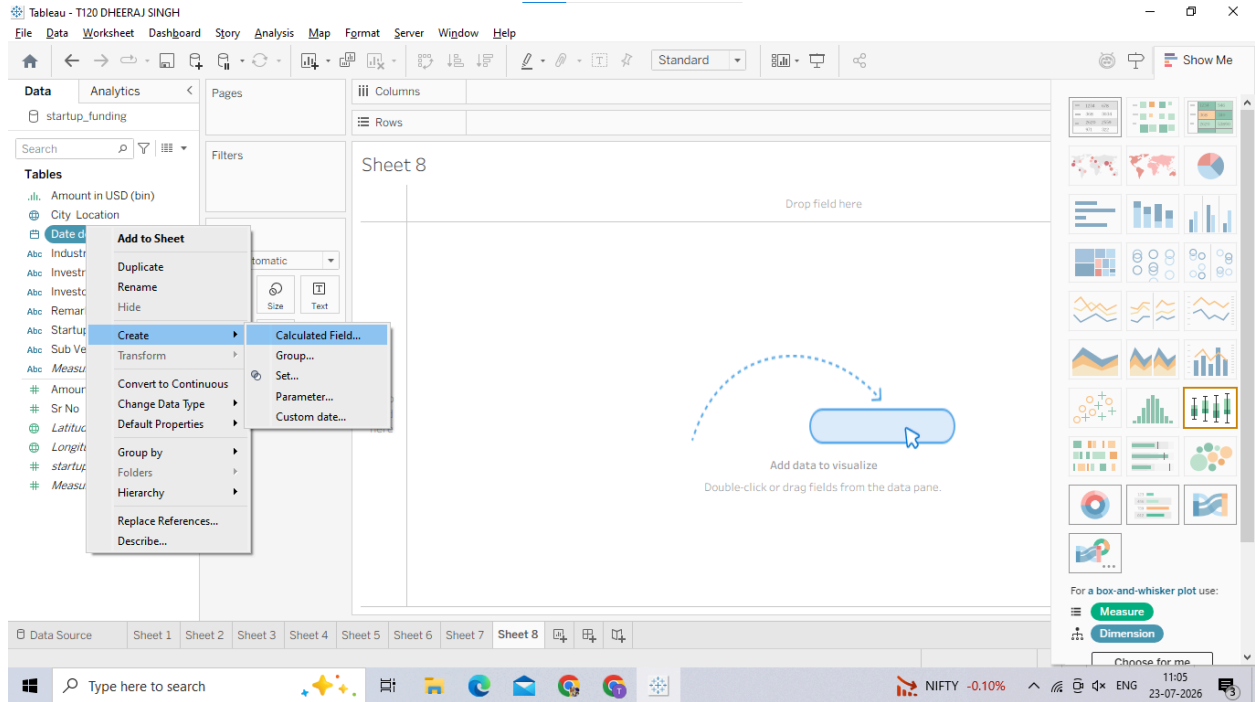
### 1. Understanding Gantt Charts in Tableau

A Gantt Chart is traditionally used to show project schedules and the duration of activities over time. In a standard Gantt chart, the horizontal axis (X-axis) represents a continuous timeline, marks are displayed as horizontal bars, and the length of each bar represents the duration of that specific event.

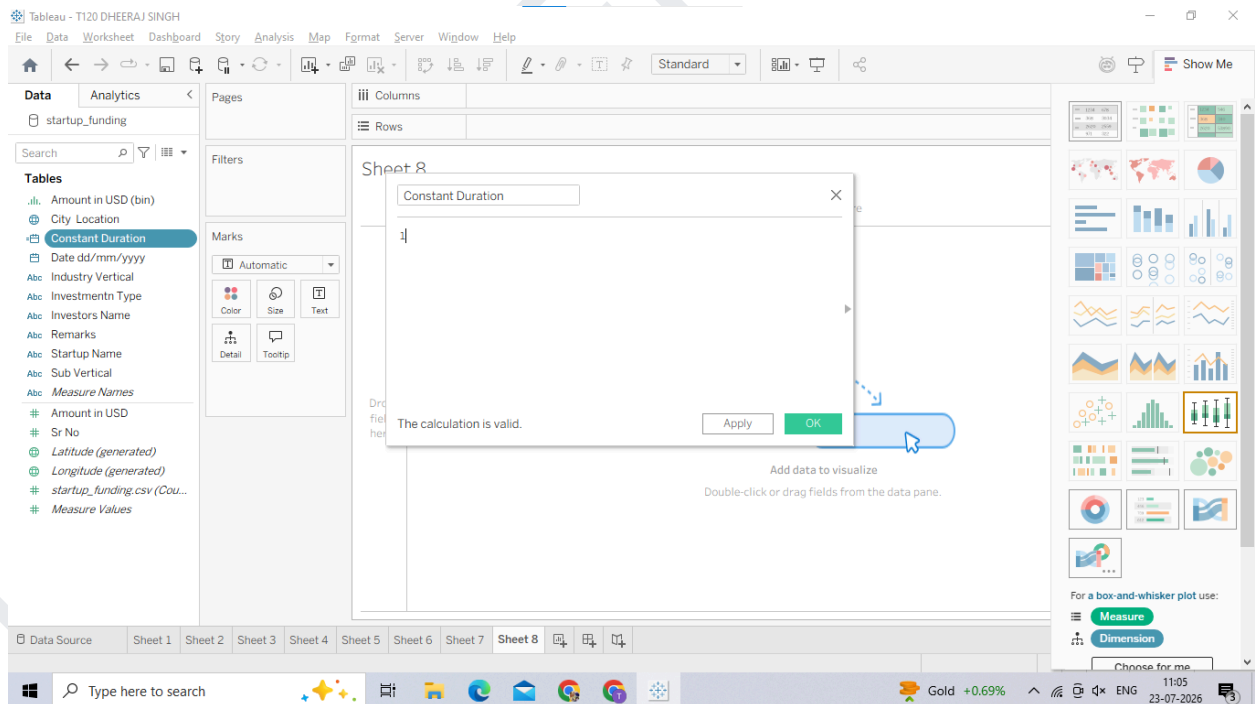
To build a meaningful duration layout using our funding dataset, we will calculate the time spread across funding dates or analyze milestone timelines. However, Tableau also allows us to use the Gantt Bar mark type to build advanced tracking timelines. Let's create an advanced chronological timeline of funding events.

### 2. Building an Advanced Date Gantt Timeline

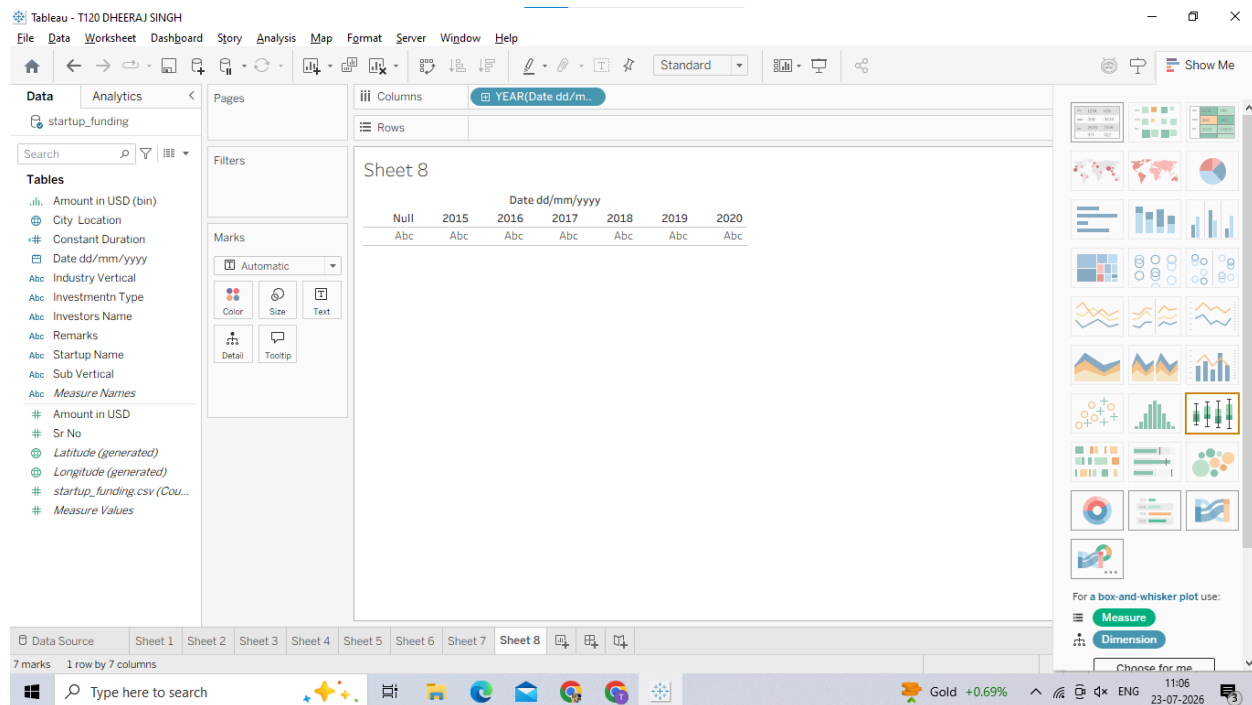
1. Click the New Worksheet icon at the bottom of the screen to open a fresh canvas (Sheet 8).
2. Look at your left Data Pane. Right-click on your Date dd/mm/yyyy field and select Create > Calculated Field.



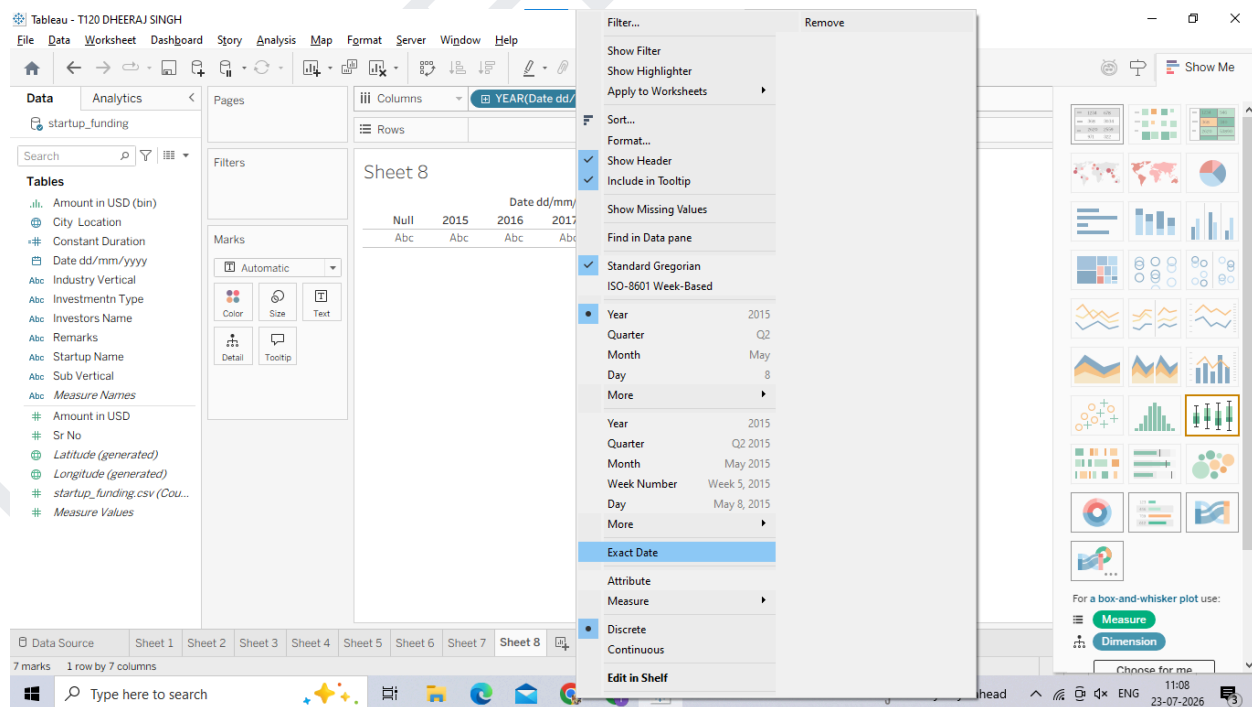
3. Name this new calculation Constant Duration. In the calculation box, type simply: 1 (This acts as a uniform placeholder length for our timeline blocks). Click OK.



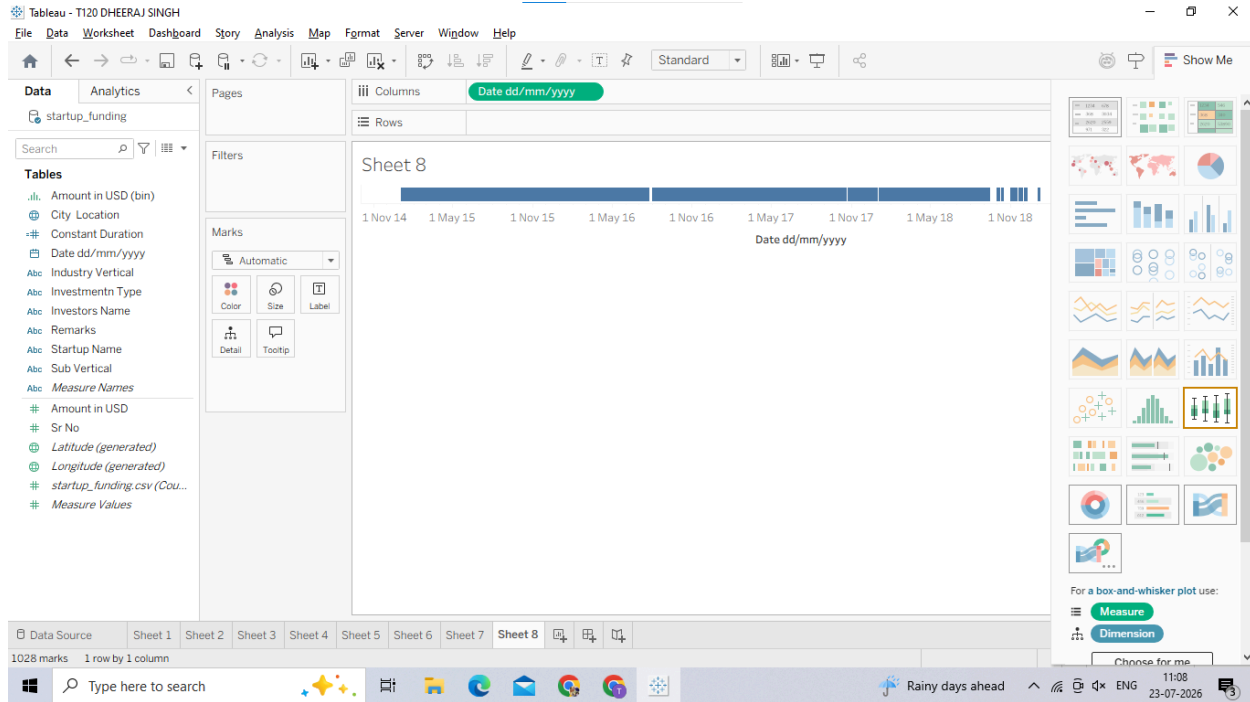
#### 4. Drag your continuous Date dd/mm/yyyy field and drop it onto the Columns shelf.



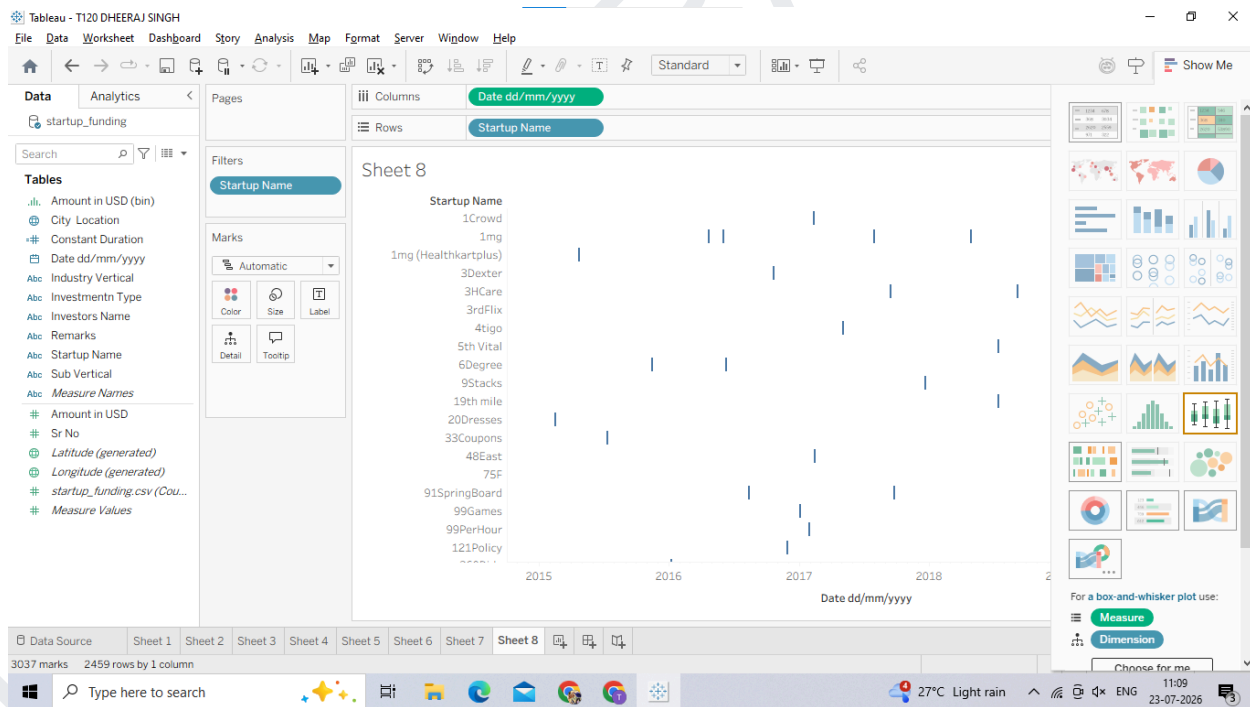
o Adjustment: If Tableau groups it automatically as a discrete year pill (blue YEAR(Date) pill), right-click that pill on the Columns shelf and select the continuous Exact Date option from the middle section of the dropdown menu. The pill color will change to green.



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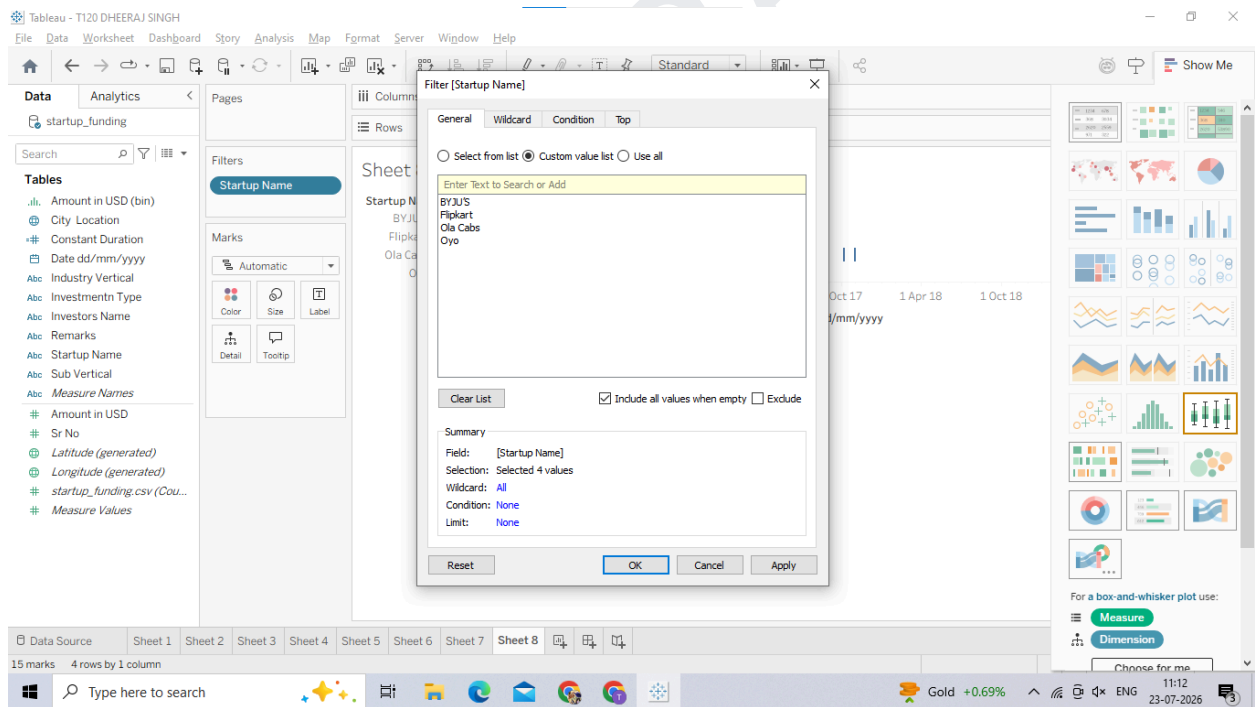
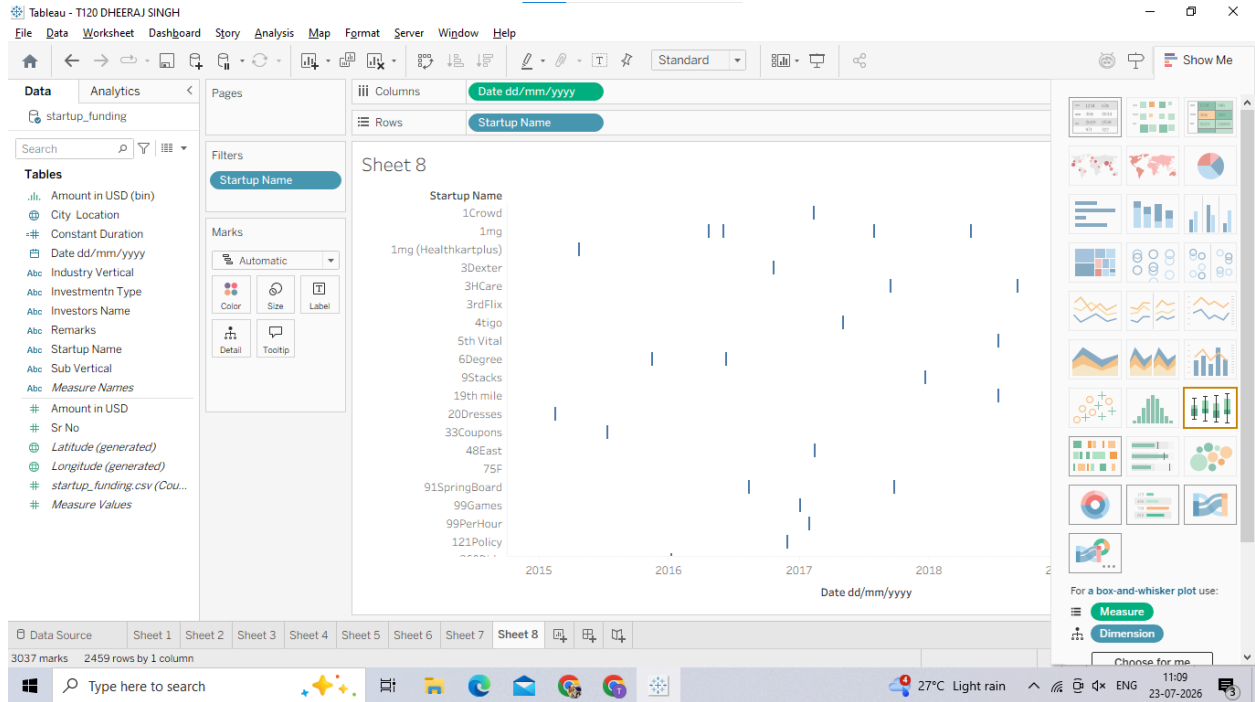


5. Drag your Startup Name dimension from the Data Pane and drop it onto the Rows shelf.



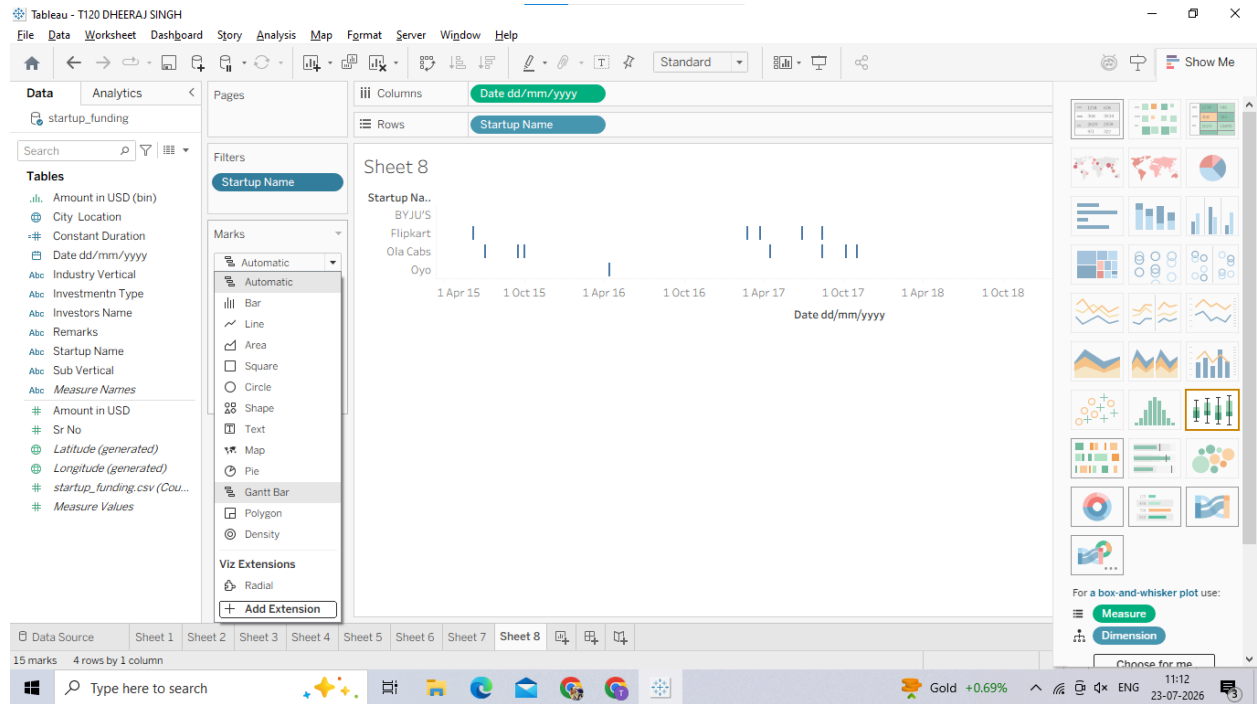
6. Apply the Filter: Because there are thousands of startups, let's look at a focused subset. Drag Startup Name into your Filters shelf, select a few notable Indian tech giants (e.g., BYJU'S, Flipkart, Ola Cabs, Zomato), and hit OK.

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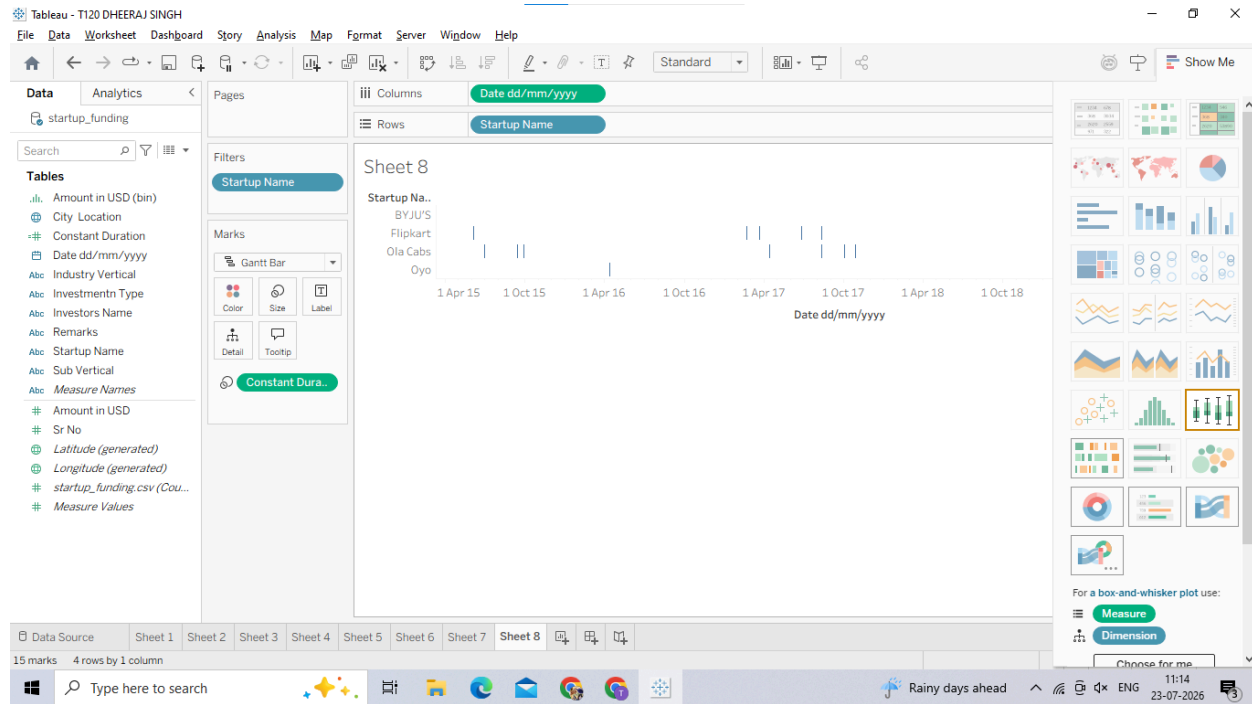


7. Look at your Marks Card dropdown menu (which currently says Automatic). Click it and manually switch the mark style option to Gantt Bar.

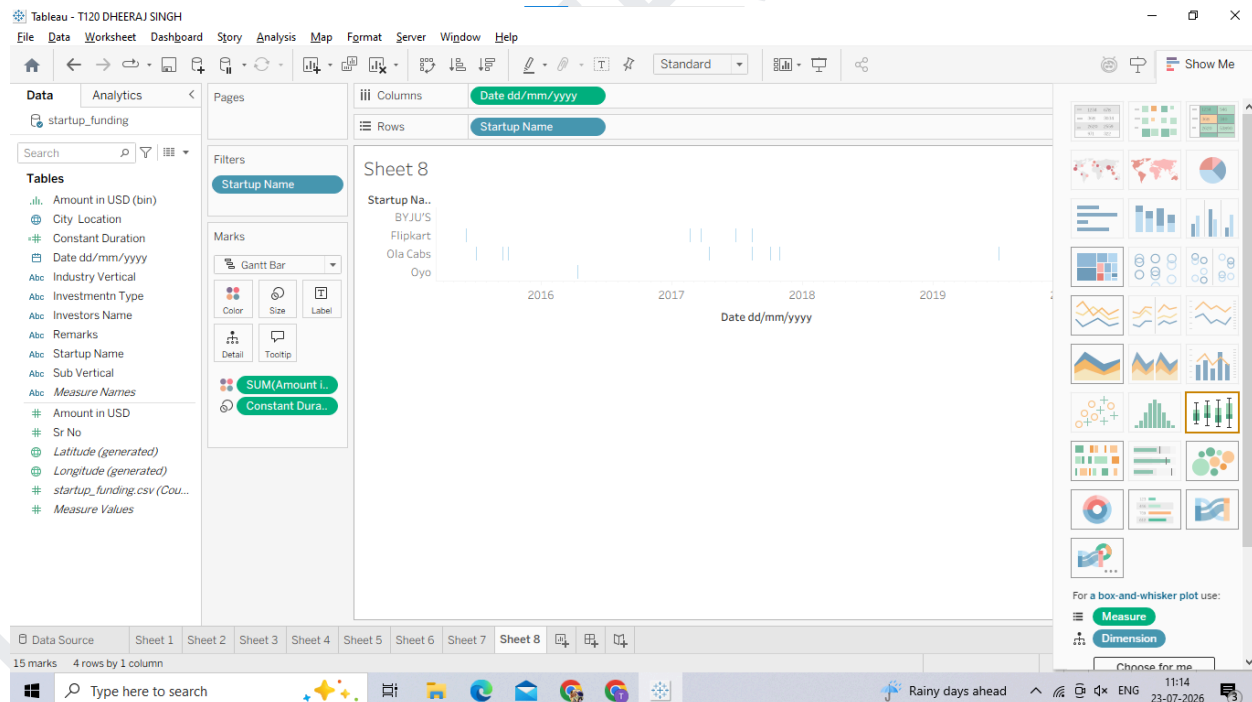
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8. Take your newly created calculated field Constant Duration from the Data Pane and drag it directly onto the Size tile inside your Marks card.



9. To make the timeline highly informative, drag your quantitative measure Amount in USD from the Data Pane and drop it directly onto the Color tile in the Marks card.



## **Why we are doing it**

A Gantt Chart/Advanced Date Visualization maps distinct historical milestones along a linear, continuous time continuum. By forcing the mark type to a Gantt Bar, assigning a standardized size duration block, and overlaying a funding color scale, we map out the exact points in history when capital injections hit these specific firms.

## **Usage**

Use Gantt layouts and advanced date visualizations when tracking project phases, product release cycles, employee operational timelines, or corporate growth histories. In our Indian Venture Capital ecosystem scenario, it creates a visual ledger—showing you exactly when top startups raised capital over the years, who went through intense, rapid clusters of concurrent funding rounds, and how their relative deal scale evolved over time.

This officially concludes Lesson 3! You have successfully mastered intermediate and advanced visualization frameworks including multi-category comparison bars, nested area treemaps, coordinate scatter plots, dual-axis scales, distribution density histograms, quartile box plots, and continuous chronological timeline Gantt charts.